STRUCTURAL ANALYSIS AND DESIGN
OF
EARTHQUAKE RESISTANT BUILDING

STRUCTURALREPORT

Owner: MR. RAJENDRA ARYAL

Submitted to:

Shankharapur Municipality-9, Kathmandu

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DECEMBER 2025

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1 INTRODUCTION

The purpose of this report is to present general overview on structural analysis & design calculation of the **two and half storey** residential Building at Kathmandu owned by **Mr. Rajendra Aryal.** This report presents the criteria and methodology that have been used throughout in structural calculation which forms a basis for structural analysis and design regarding the assumption of loadings, follow up of the standards as well as the for relevant methodologies.

2 DESIGN PHILOSOPHY

The past devastating earthquakes have proved the vulnerability of most of the vernacular buildings of Nepal. Enormous life and property were lost due to the collapse of buildings Earthquakes can neither be prevented nor predicted precisely. But large- scale destruction can be minimized by employing seismic-resistant measures in buildings. This can be achieved by the use of existing building materials in appropriate analysis.

The structures of buildings are designed to provide the spatial requirements in accordance with the purpose of the buildings taking into account the aesthetic aspect and provision for various services and systems necessary for the operation of airport. In order to fulfill such requirements, the structural materials and the structural system have been determined taking into account the availability of materials, cost efficiency and structural integrity, i.e. stability, strength and serviceability; the three main factors to be in corporated in the design of all structures.

3 DESIGN CODES AND STANDARDS

The structural design is based primarily on the current National Building Codes and Standards of India, while the relevant British Standards, European Standards or National Building Codes of Nepal are referred to appropriately for the areas/provisions that are addressed in the Indian Standards or as required for the purpose of design. The main design standards followed for structural design are given below, indicating their area of application.

S.N.	Codes and Standards	Description
1	IS 456: 2000	Plain and Reinforced Concrete - Code of Practice
2	IS 875 (Part 1): 1987	Code of Practice for Design Loads (other than Earthquake) for Buildings and Structures: Part 1 Dead Loads – Unit Weights of Building Material and Stored Materials (Second Revision)
3	IS 875 (Part 2): 1987	Code of Practice for Design Loads (other than Earthquake) for Buildings and Structures
4	IS 875 (Part 3): 1987	Code of Practice for Design Loads (other than Earthquake) for Buildings and Structures
5	SP 34: 1987	Handbook on Concrete Reinforcement and Detailing
6	IS 13920: 1993	Ductile detailing of reinforced concrete structures subjected to seismic forces - Code of practice
7	IS 1893: 2016	Criteria for Earthquake Resistant Design of structures
8	IS 383: 1970	Specification for coarse and fine aggregates from natural sources
9	IS 1786: 1985	Specification for high strength deformed steel bars and wires for concrete reinforcement (superseding IS:1139 - 1966)
10	IS 12269: 1987	Specification for 53 Grade Ordinary Portland Cement.
11	IS 1904: 1986	Design and Construction of Foundation in soils: General Requirements.
12	IS 800: 2007	Code of Practice for General Construction in Steel (Third Revision)
13	NBC 104: 1994	Wind Load
14	NBC 105: 2077	Seismic Design of Buildings in Nepal
15	NBC 205: 2024	Ready-To-Use Detailing Guideline for Low Rise Reinforced Concrete Buildings Without Masonry Infill
16	NBC 109: 1994	Masonry: Unreinforced
17	NBC 202: 1994	Mandatory Rules of Thumb Load Bearing Masonry
18	EN 1998 - 1	Eurocode 8: Design of structures for earthquake resistance - Part 1: General rules, seismic actions and rules for buildings

4 GENERAL DESIGN CRITERIA

4.1 MATERIALS

The materials to be used in structural design are as follows.

Concrete

Cast-in-situ concrete for all structural elements:

For Slabs, Beams	-	M20
Columns,	-	M25
For Footings, Lintel/Earthquake Bands	-	M20

Reinforcing Bars

High Strength deformed steel bars: Grade Fe 500 to 1786 (14% minimum elongation)

4.2 SOIL BEARING CAPACITY

In absence of soil test report, the presumed bearing capacity of foundation for the site considering combined footing at 2 m depth is **120 KN/m^2**.

4.3 METHODOLOGY

ETABS version 20 has been used for the linear static and dynamic analysis and design of three- dimensional structures, in which the spatial distribution of the mass and stiffness of the structure is adequate for the calculation of the significant features of structures. The structural elements of reinforced concrete were designed to Limit State Theory, while the structural steel elements have been designed to the Permissible Stress Theory. The major structural elements were designed using design spreadsheets/software and verified manually too following the design aids SP16, IS 456 and IS 800 for reinforced concrete structures and structural steel elements respectively.

Structures designed and built in seismic regions shall fulfill following fundamental requirements ie. Life safety and damage limitation. For the verification of performance requirements, following limit state shall be checked.

1. Ultimate Limit State (ULS)

   Ultimate limit states are associated with collapse or with other forms of structural failure which might endanger the safety of people. Design for ultimate limit state represents a procedure that ensures the probability of collapse of a structure is at an acceptable level.

2. Serviceability Limit State(SLS)

   Damage limitation states are associated with damage beyond which specified service requirements are no longer met. It represents a level of force within the structure below which there is a high degree of assurance that the structure can continue to be used as originally intended without repair.

Design loads for all structures have been in accordanc to the criteria described below.

5 DESIGN LOAD

5.1 DEAD LOAD

Dead Load on the structure comprise the self-weight of the member; weight of the finishes and partition walls. These are usually dependent upon the constructional features and have to be assumed in order to design various structural concrete members. The Loads on the beams due to the slabs were calculated according to clause 23.5 of IS 456 – 2000. The Wall Load is taken for thickness of either 230 mm or 115 mm as per Architectural Drawing and suitable reduction is made for Window and Door Opening.

All loads/forces due to gravity on the components of the building structure including the structures self-weight, roofing, flooring, suspended ceiling, wall/partition, services including machinery, piping, rack with all associated finishing permanently attached thereto are calculated in accordance with IS: 875 (Part 1) -1987.

As the software, we have used, generates the self-weight of the Structural member by itself, we have not calculated the self-weight.

The structure has been designed considering following ideal loads:

Unit Weight (Material)

a. Density of Concrete = 25kN/m3
b. Live Load = 0.75 kN/m2 for Not accessible Floor
   = 1.5 kN/m2 for Accessible Floor
   = 3.00 kN/m2 for Staircase and Corridors
   =2.0 kN/m2 for rooms
c. Floor Finishing Load = 1.2kN/m2
d. Density of Brick Wall =19.2kN/m^2
e. Soil Bearing Capacity (adopted)= 120 kN/m^2
f. Dead loads were considered separately as;

Self-weight	=	DL
Wall Load	=	DL
Floor Finishing	=	DL

5.2 IMPOSED LOAD

The load assumed to be produced by the intended use and occupancy of the building, including the loads of movable partition, impact, vibration, but excluding wind, seismic, snow and other loads due to temperature changes, creep, shrinkage, differential settlement, etc., in accordance with IS 875 (Part2) – 1987. The words, “liveload”,here in or elsewhere shall be readas “imposedload”as synonymous.

Wall Load	(Assuming 30 % opening)					
Wall load calculation	**B`**	**Density**	**H**	**Without opening**	**With opening**	**Unit**
9" wall(on grid)	0.23	19.2	2.39	10.55	7.39	KN/m
Plaster 12.5mm	0.013	20.4	2.39	0.61	0.43	KN/m
Total				11.16	7.81	KN/m
Adopted value				**11.5**	**8**	KN/m
4" wall (on grid)	0.115	19.2	2.39	5.28	3.96	KN/m
Plaster 12.5mm	0.013	20.4	2.39	0.61	0.43	KN/m
Total				5.89	4.38	KN/m
Adopted value				**6**	**4.5**	KN/m
Parapet wall						
4" wall (on grid)	0.115	19.2	1.07	2.36		KN/m
Plaster 12.5mm	0.013	20.4	1.07	0.27		KN/m
Total				2.63		KN/m
Adopted value				**2.7**		KN/m

5.3 EARTHQUAKELOAD

Response Spectrum including modal analysis hasbeen used as per theNBC 105:2020 tocalculate theearthquake forces on the structure. Base shear was derivedfrom the horizontal seismic base shear (Vb) given by:

$V=C_d(T_1)W$ (NBC 105:2020, Cl. 6.2(1))

Where;

W= Seismic weightof thestructure equalto thetotal dead loadplusappropriate amounts of specified imposedload

$C_d(T_1)$ = Horizontal base shear coefficient given by:

a) Ultimate Limit State

$$C_d(T_1) = \frac{C(T_1)}{R\mu \times \Omega u}$$

(NBC 105:2020, Cl.6.2(1))

Where,

Elastic Site Spectra($C(T1)$) is given by

$$C(T) = C_h(T)\, Z\, I.$$

(NBC 105:2020, Cl.4.1.1)

Where,

$C_h(T)$ = 2.25 Spectral Shape Factor (NBC 105:2020, Cl.4.1.2)

I = 1.0Importantfactor(NBC 105:2020, Cl.4.1.5)

Z = 0.35Seismic Zoning Factor(KATHMANDU) (NBC 105:2020, Cl.4.1.4)

So,C(T)=2.25*0.35*1.0

=0.7875

R□= The ductility factor=4 (NBC 105:2020, Cl.5.3.1 table 5-2)

□u=The overstrength factor=1.5 (NBC 105:2020, Cl.5.4.1 table 5-2)

$C_d(T_1)$ = 0.7875/(4*1.5)

= **0.1313**

b) Serviceability Limit State

$$C_d(T_1) = \frac{Cs(T_1)}{\Omega s}\ldots$$

(NBC 105:2020, Cl.6.2(2))

Where,

$C_s(T)$ = 0.2C(T)Elastic Site Spectra (NBC 105:2020, Cl.4.1.2)

= 0.2*0.7875

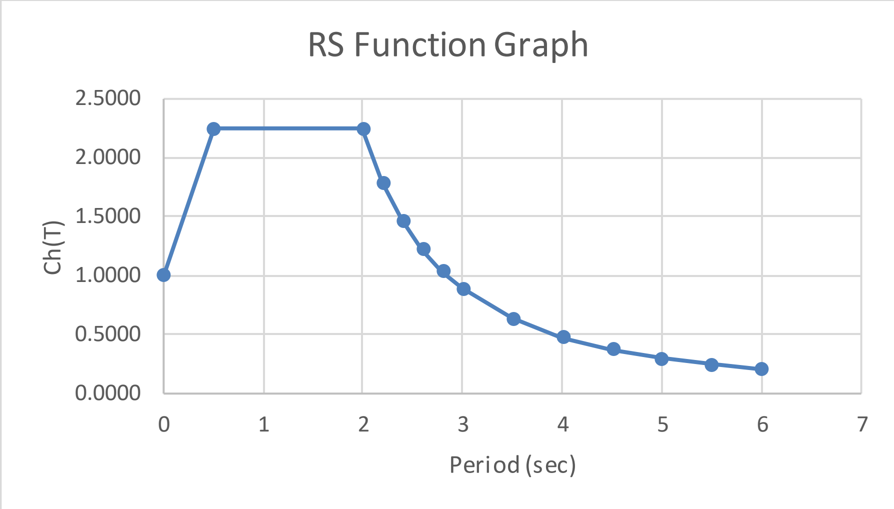
= 0.1575

□□□□□s= The overstrength factor=1.25 (NBC 105:2020, Cl.5.4.2 table 5-2)

So,

$C_d(T\ 1)$=0.1575/1.25

=**0.126**



Period of vibration

The fundamental translation period shall be determined by the following two methods.

1. Rayleigh Method
2. Emperical Equations

The time period obtained by empirical equations shall be compared with that obtained by Rayleigh Method and the lesser value of the two shall be adopted for the design action.

Emperical Equations

The approximate fundamental natural period of vibration (Ta), inseconds, of a moment resisting frame building without brick infill panels may be estimated by the empirical expression:

$Ta = 0.075\ h^{0.75}$ for RCMRF building

$= 0.085\ h^{0.75}$ for steel MRF building

Where:

h =Height of building from foundation, in m.

$T_1 = 0.075 * 8.3544^{0.75}$

$=0.375$sec.

The approximate fundamental time period as per NBC 105: 2020 CL: 5.1.3 (Page 29)

$T_1 = 1.25*0.53$

$= 0.468$ sec

Rayleigh Method

$$T_1 = 2\pi \sqrt{\frac{\sum_{i=1}^{n}(W_i d_i^2)}{g\sum_{i=1}^{n}(F_i d_i)}} \quad \text{..................................5.1(1)}$$

Where

d_i = elastic horizontal displacement of center of mass at level i, ignoring the effects of torsion.

F_i = lateral force acting at level i

g = acceleration due to gravity

i = level under consideration

n = number of levels in the structure

W_i = seismic weight at level i

The seismic weight at each level, *Wi*, shall be taken as the sum of the dead loads and the factored seismic live loads between the mid-heights of adjacent stories.
The seismic live load shall be determined as given in **Table 5-1**.

***Table 5-1**: Live Load Categories and Factors*

Live Load Category	Factor (λ)
Storage	0.60
For Other Purpose	0.30
Roof	Nil

6 LOAD COMBINATION

For design of reinforced concrete elements, the following load combinations were considered according to NBC 105:2020 CL 3.6.1:

1. 1.2DL+1.5LL
2. DL+0.3LL+SLSEQX
3. DL+0.3LL-SLSEQX
4. DL+0.3LL+SLSEQY
5. DL+0.3LL-SLSEQY
6. DL+0.3LL+ULSEQX
7. DL+0.3LL-ULSEQX
8. DL+0.3LL+ULSEQY
9. DL+0.3LL-ULSEQY

Where:
DL=DeadLoad

LL= LiveLoad

ULSEQX= Ultimate limit state Earthquake load inX-direction

ULSEQY= Ultimate limit state Earthquake load inY-direction

SLSEQX= Serviceability limit state Earthquake load inX-direction

SLSEQY= Serviceability limit state Earthquake load inY-direction

7 STRUCTURAL ANALYSIS USING ETABS

The building was modeled in structural analysis program ETABS Ver. 20 using frame elements. Slabs were modeled using shell elements to distribute loads on frame elements.3-D Modeling was prepared using this FEM modelingbased software considering beams and columns as FRAME element, and Slab asshell element wherever possible.

Equivalent Static Method was carried out for its analysis in lateral loading due toearthquake

8 DESIGN METHODS OF STRUCTURAL ELEMENTS

We have followed Indian Standard Codeof Practicefor Plainand Reinforced Concrete IS: 456-2000 for design of Structural Elements. This incorporates the two methods ofStructural Design of RC structures specified as:

a. Working Stress Method based on the Working loads in conjunction with permissible stresses in thematerials.
b. Limit State Method based on safety and serviceability requirements associated with the design loads and design strengths of the materials. These design loads and design strengths are obtained by applying partial safety factors for characteristic loads and strengths of the materials concrete andsteel.

We have followed the limit state method which is incorporated in IS:456-2000. It is consistent with the new philosophy of design termed limit state approach which was incorporated in the Russian Code – 1954, the British code BS 8110 – 1985 and the American Code ACI 318 – 1989.

9 LIMIT STATE METHOD

- **LimitStates**

The Limit State method of design covers the various forms of failure. There are several limit state at which the structure ceasesto function, themost importantamong thembeing,

a. The limit state of collapse or total failure ofstructure.

It correspondsto themaximum loadcarrying capacity. Violation ofcollapse implies failure. This limit state corresponds to Flexure, Compression, Shear and Torsion.

b. The limit state of serviceability which includes excessive deflection and excessive local damage. Excessive deflection adverselyaffects thefinishes and excessive local damage results in cracking of concrete, which impairs the efficiency, or appearance of the structure.

- DesignEquationsinLimitStateMethod:

a) Singly ReinforcedSections:

The depth of Neutral Axis xu is obtained as Xu= [(0.87Astfy / 0.36 fck b)]

For under Reinforced Sections, the neutral Axis depth Xu is limited to a value of Xu max < 0.48 d for steel of fy = 415 N /mm^2

If the neutral axis depth is less than these limiting values, the ultimate moment is computed

By the equation:

Mu= 0.87 fy Astd [1 – { fyAst/ b d fck }]

If the Neutral Depth is equal to the limiting values, then the section is balanced or limiting and the moment capacity is evaluated by theequation.

Mu,lim= 0.36 X u, max / d [1 – 0.42 X u,max/ d] b d ^{2} fck

Mu,lim= 0.138 fckb d ^{2} (for HYSD bars with fy = 415 N /mm^2) Area of steelina balanced section is givenby:

Ast= [0.36 f ck b X u,max/ 0.87 f y]

b) Doubly ReinforcedSections:

The ultimatemoment resistance of sections with compressionreinforcement (doubly reinforced Section) is computed by the equation.

$[M_u - M_{u,lim}] = f_{sc} A_{sc} (d - d')$

Where, M_u = Moment of Resistance of the doubly reinforced sections

$M_{u,lim}$ = Limiting moment of Resistance of a section without Compression reinforcement

d d' = Effective Depth

= Depth of Compression Reinforcement from compression face

f_{sc} = Design stress in compression reinforcement

The total Area of Tension Reinforcement is obtained as $A_{st} = A_{st1} + A_{st2}$ Where,

A_{st1} = Total Area of Tensile Reinforcement

A_{st2} = Area of Tensile Reinforcement for a singly Reinforced Section for Mu,kim

Asc = Ascfsc/ 0.87 fy

The nominal shear stress Tv in beams of uniform depth is obtained as

$T_v = V_u / b d$

Where, Vu = Ultimate shear Force due to design loads

B = Breadth of member

d = effective depth

The permissible shear stress T_c in beam depends upon the percentage of longitudinal reinforcement. Where Tc < Tv, minimum shear Reinforcement is given by

Asv / bSv> = 0.4 / fy

If Tv exceeds Tc, but isless thanTc max, then Shear Reinforcements aredesigned to take up the balanced shear Vu – Tcbd

The Strength of Shear Reinforcement Vus is calculated as Vus =0.87 fi Asv d / Sv

10 ANALYSIS AND DESIGN OF THE STRUCTURALELEMENTS:

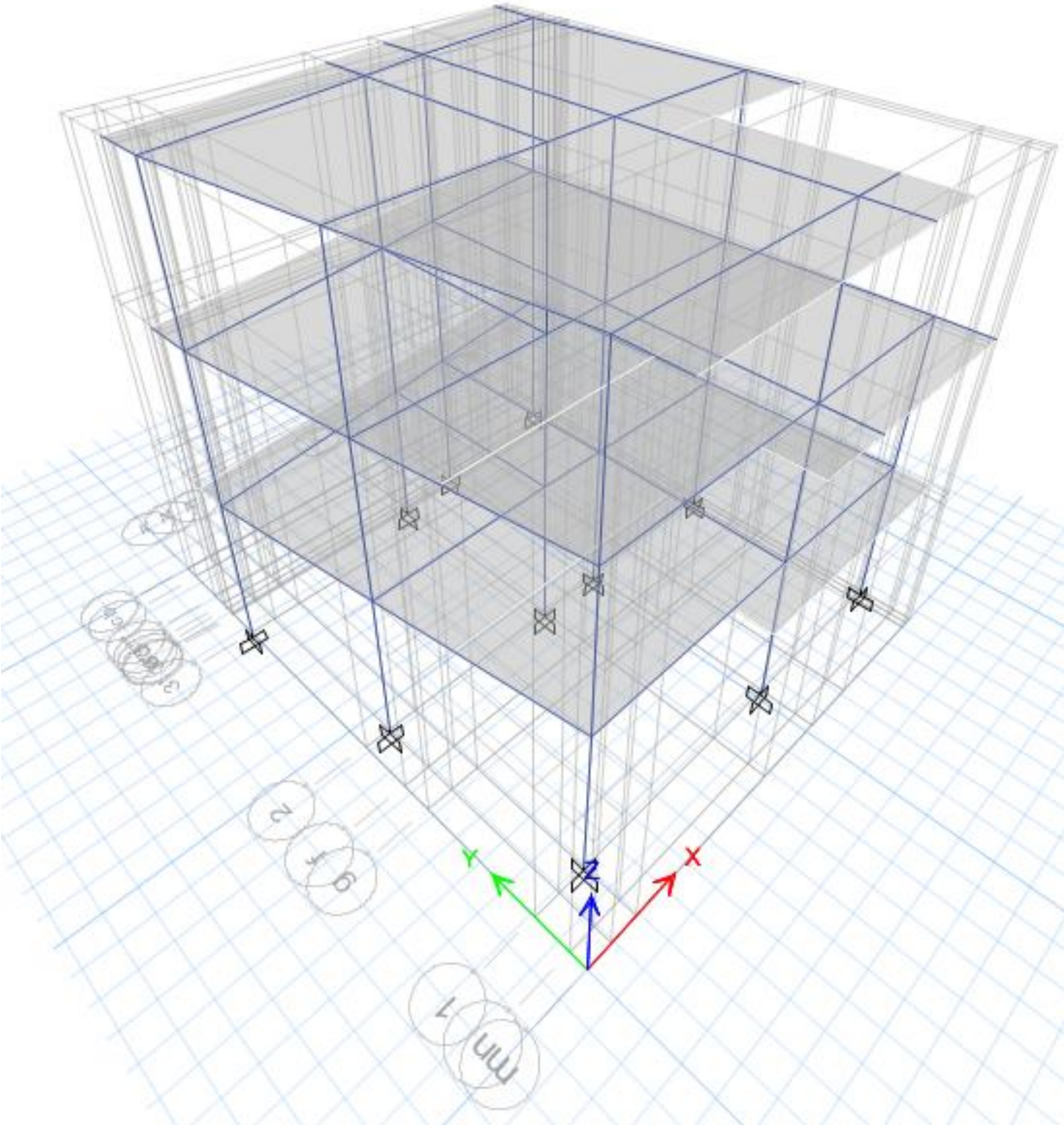
The Structure is analyzed and designed by standard software ETABS Vr.20. It is the World's most popular and widelyused structuralEngineering Software. It isequipped with the powerful analysis, design, graphics, and visualization capabilities.

General Information on Structural Elements of the Building

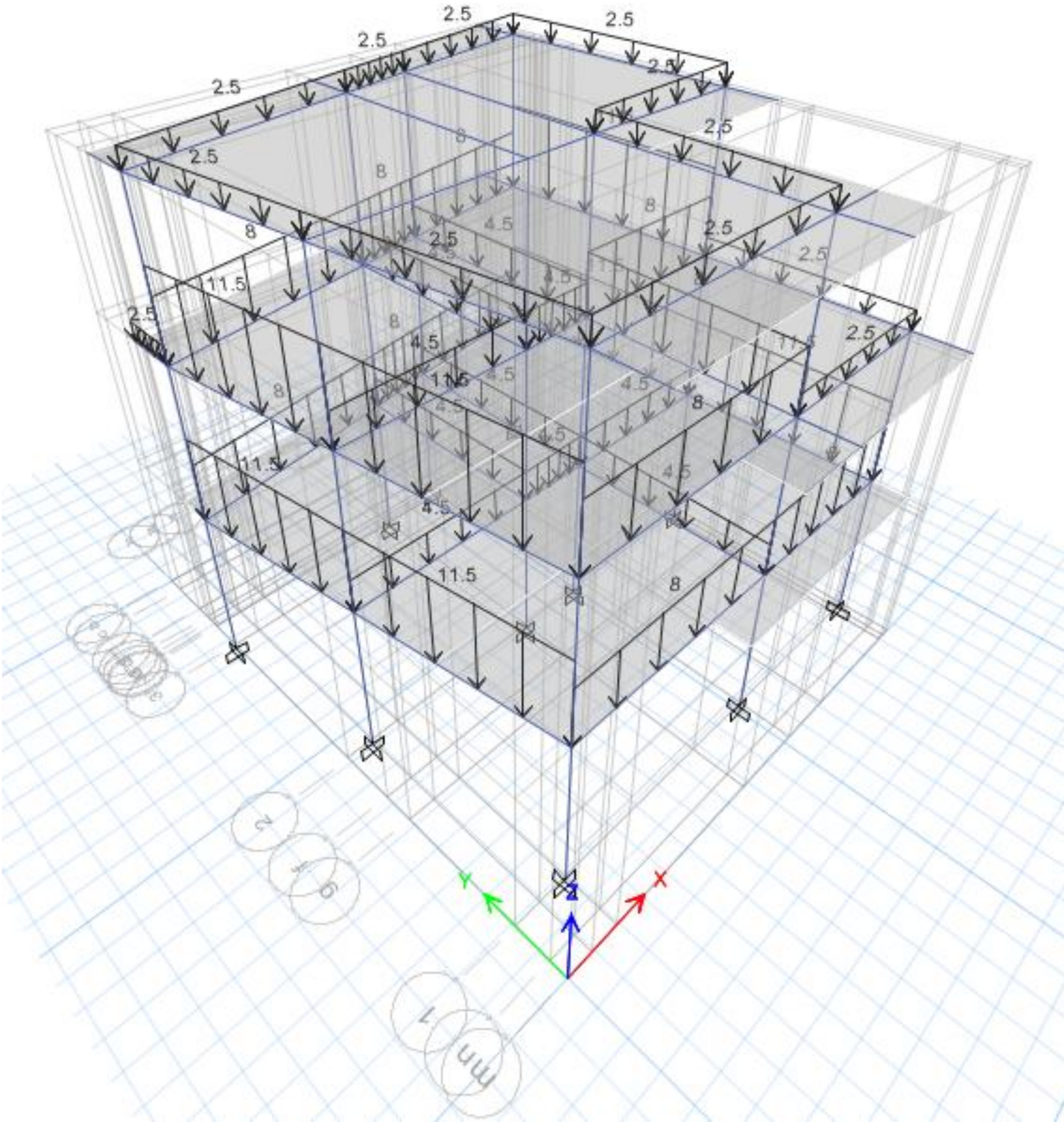
Element	Description	Grade of Concrete	Remarks
Column	350 mm X 350 mm	M20	
Main Beam	350 mm X 230 mm	M20	
Slab	125 mm	M20	
Waist Slab	125 mm	M20	
Foundation	Isolated, Combined and Strap Footing	M25	Soil Bearing Capacity is assumed to be $120 kN/m^2$

11 DETAILING OF THE STRUCTURALELEMENTS:

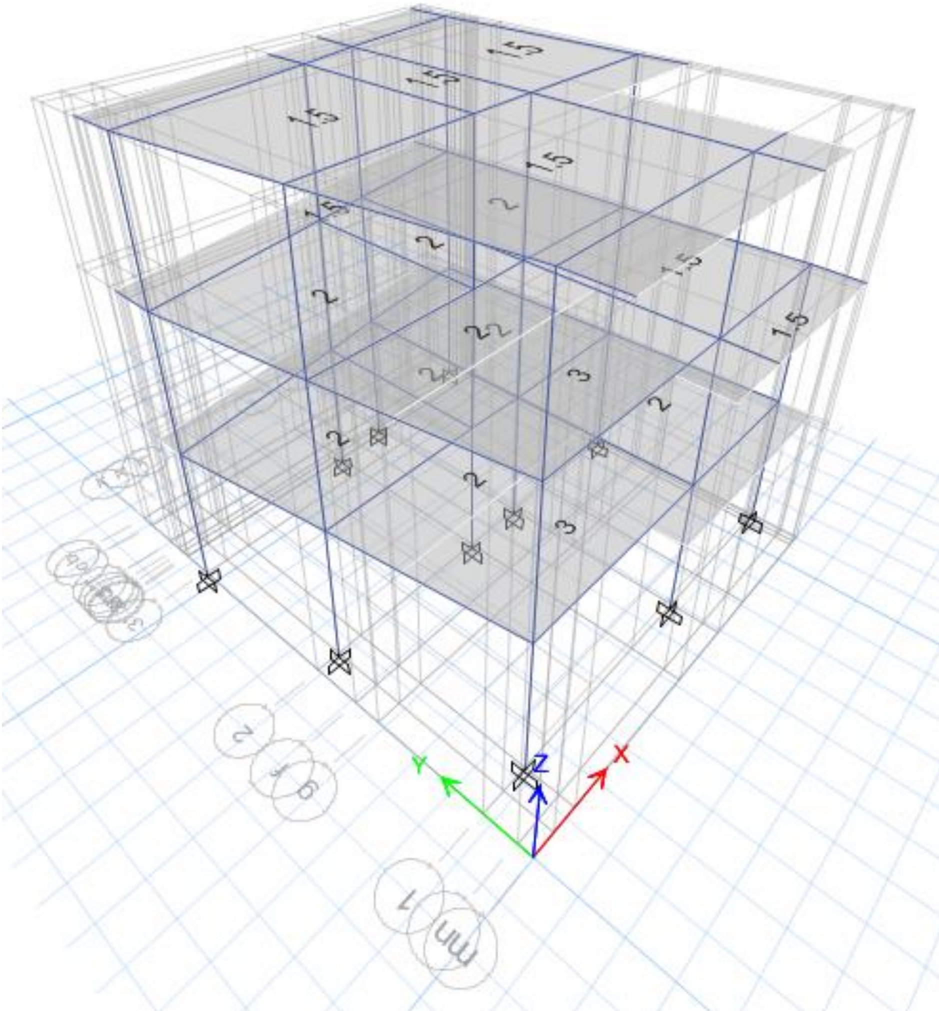
The Reinforcement detailing of most of theimportant structuralcomponents have been shown in drawing. They confirm with the relevant sections of the IS Codes IS 456-2000, IS 1893-1984, SP-16, and SP –34–1987.



Frame Structure and Element

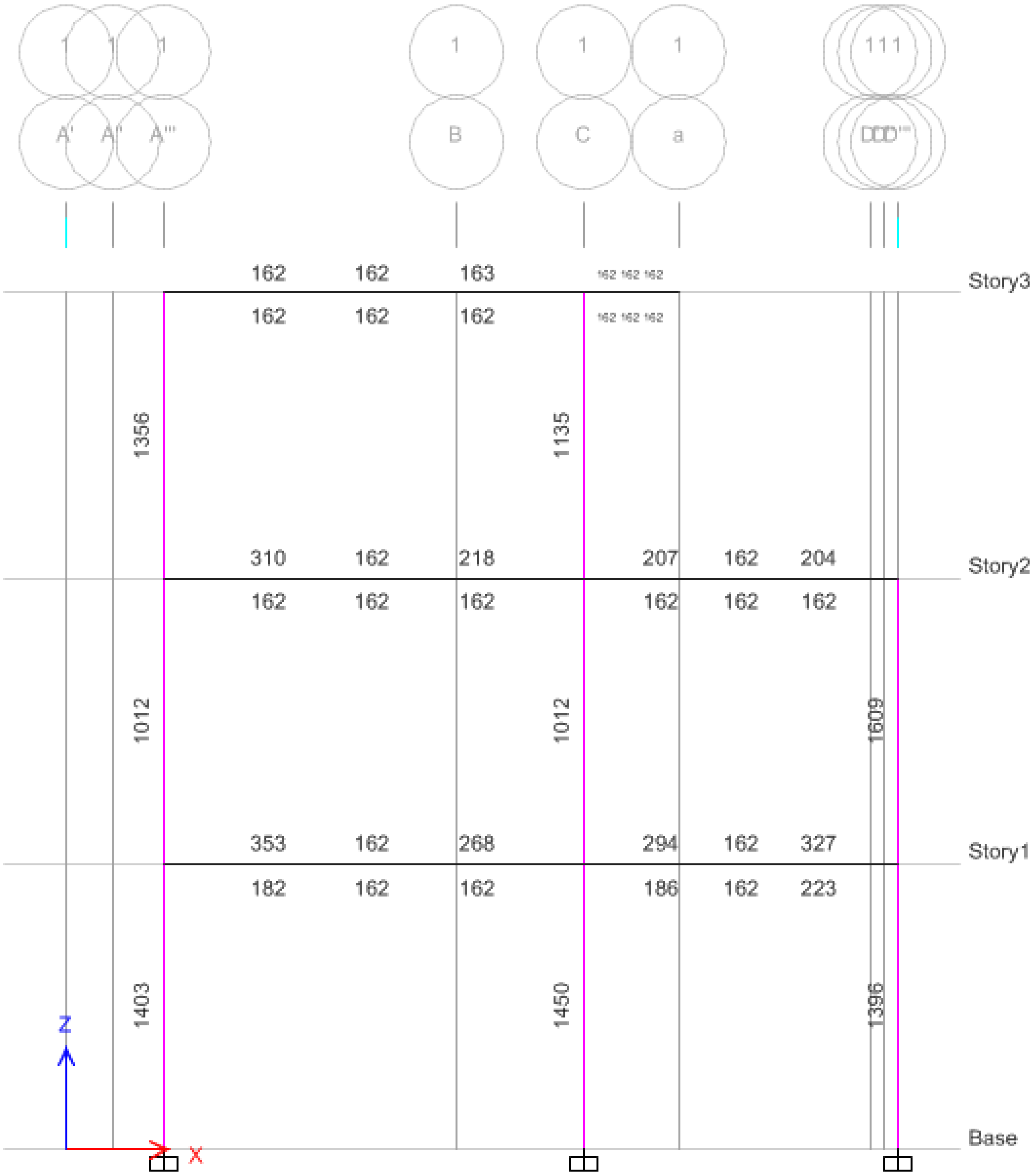


Wall Dead Load Frames (KN/m)

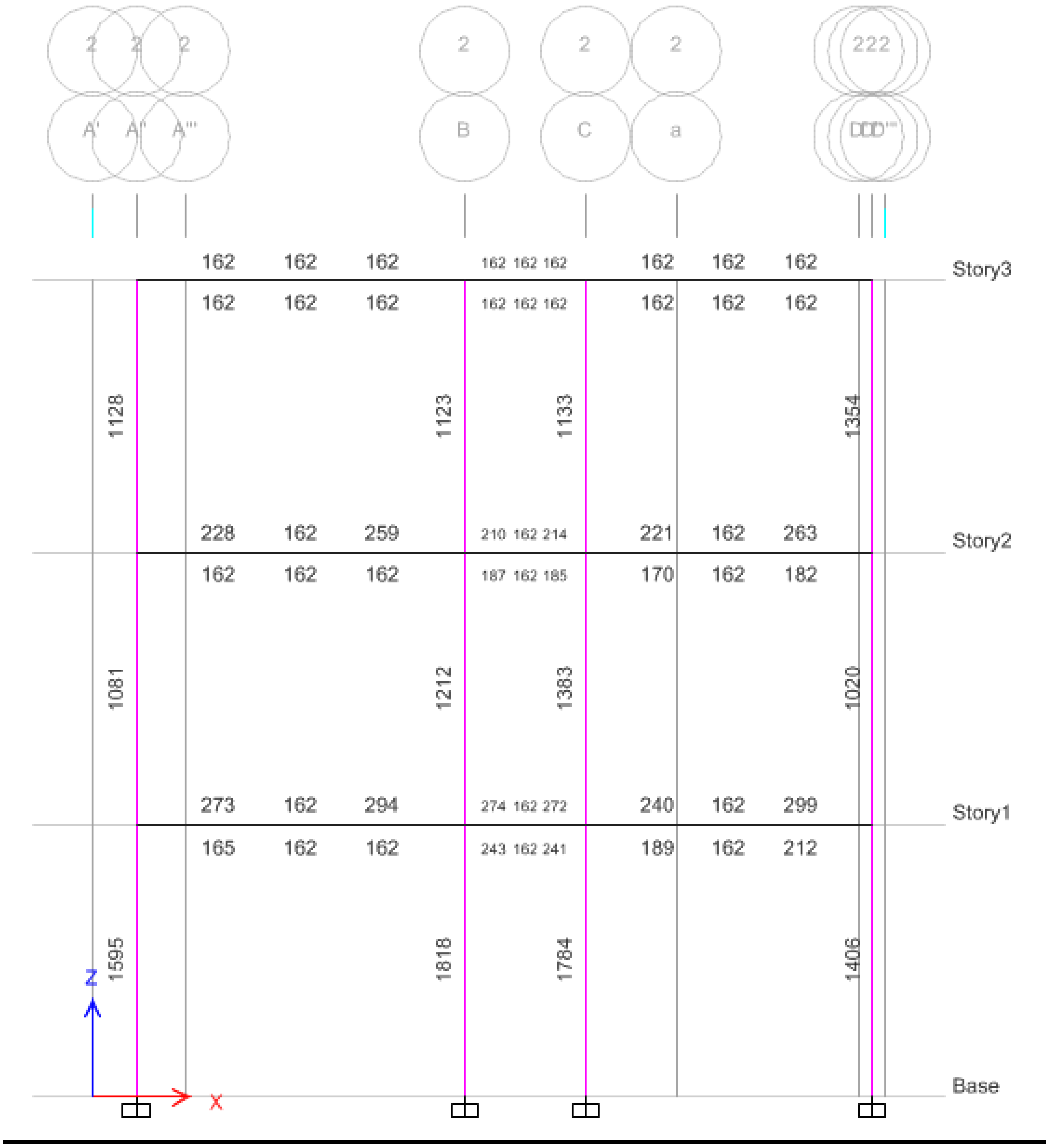


Live load (KN/m^2)

longitudinal Reinforcements (X - Z Axis) (mm^2)



Grid 1-1


Story3
Story2
Story1
Base
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1354
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162 162 162 187 162 185 170 162 182
1081
1212
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1406
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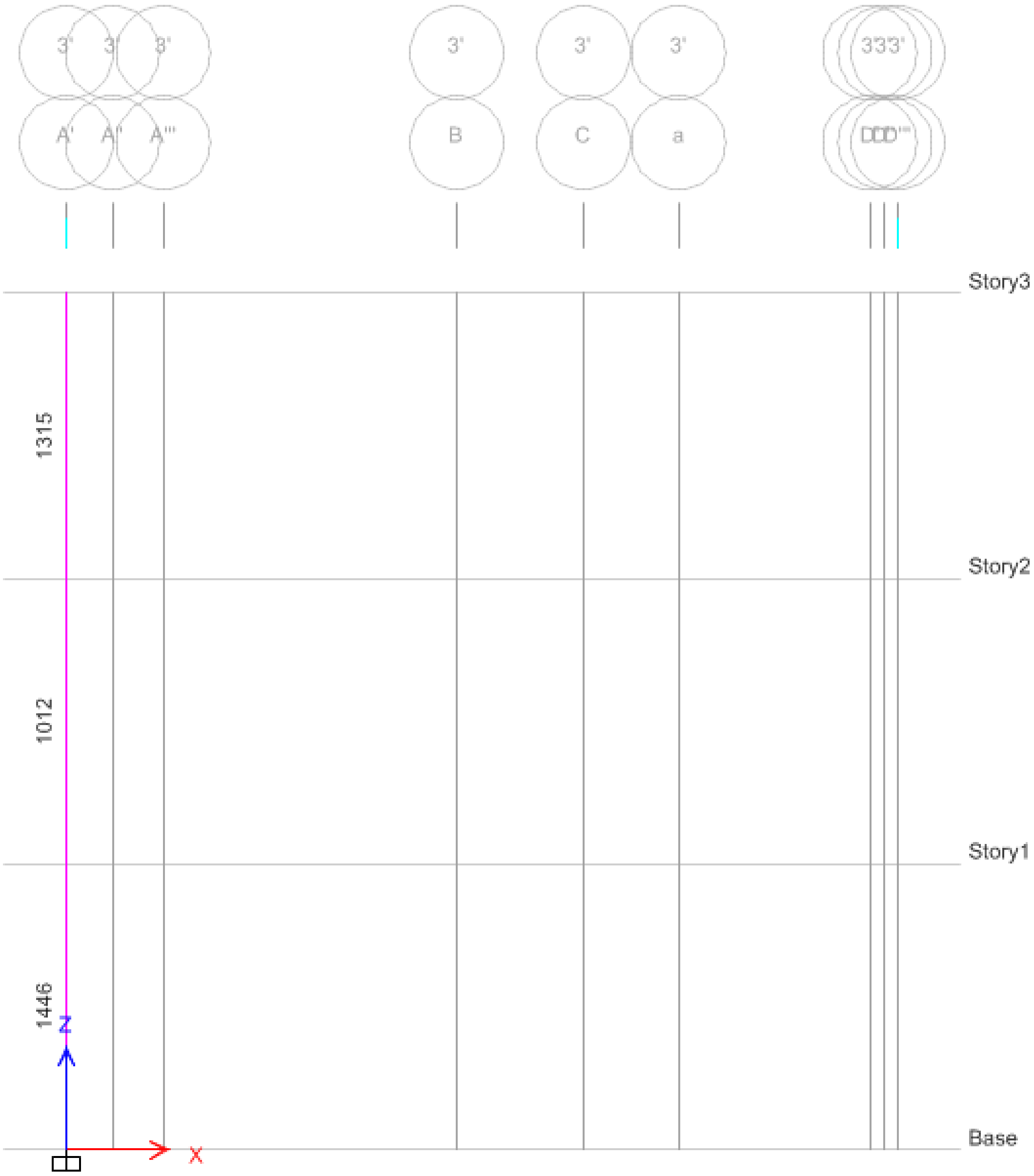
X

Grid 2-2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Story1
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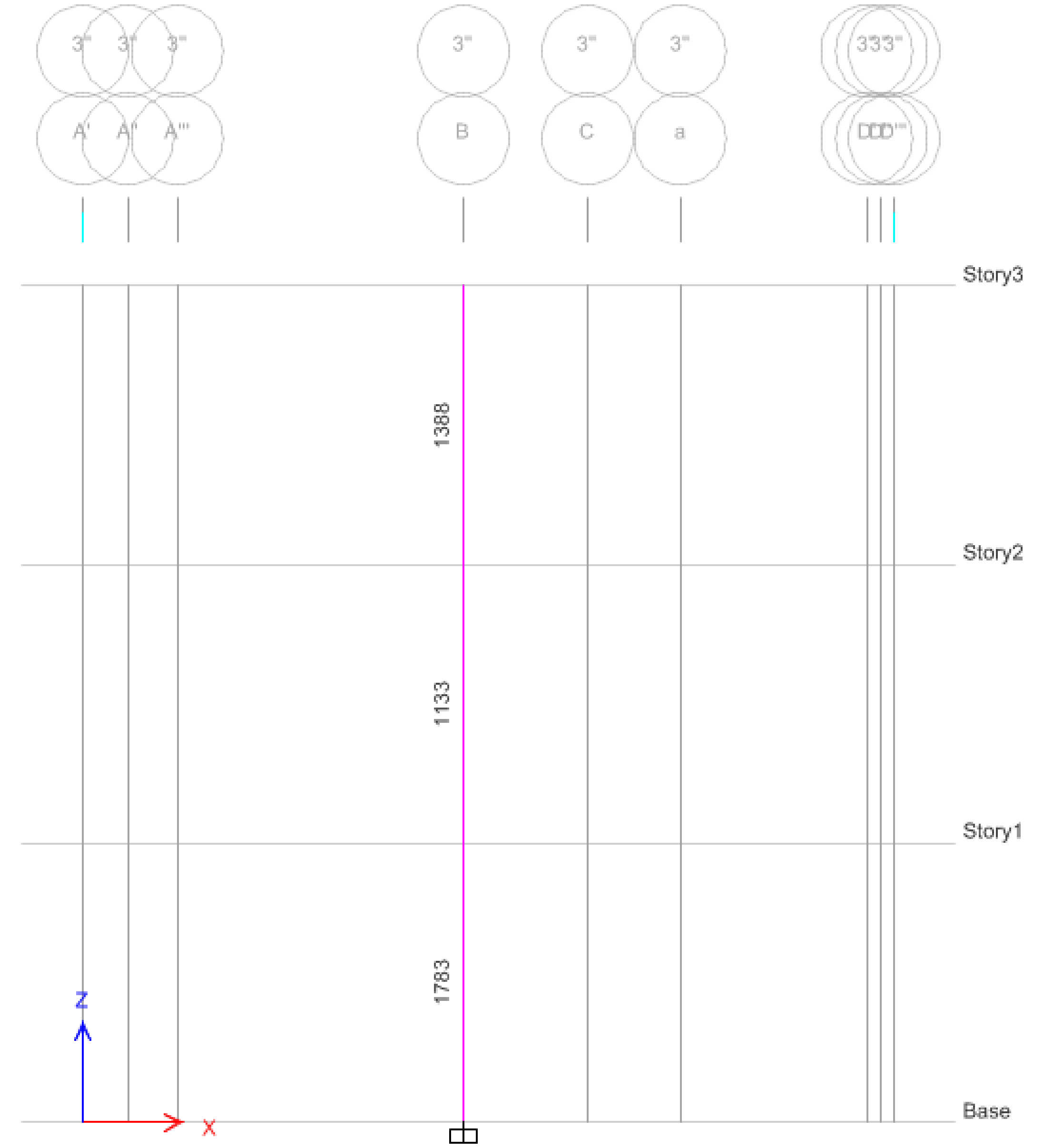
1012
1446
Z
X

Grid 3-3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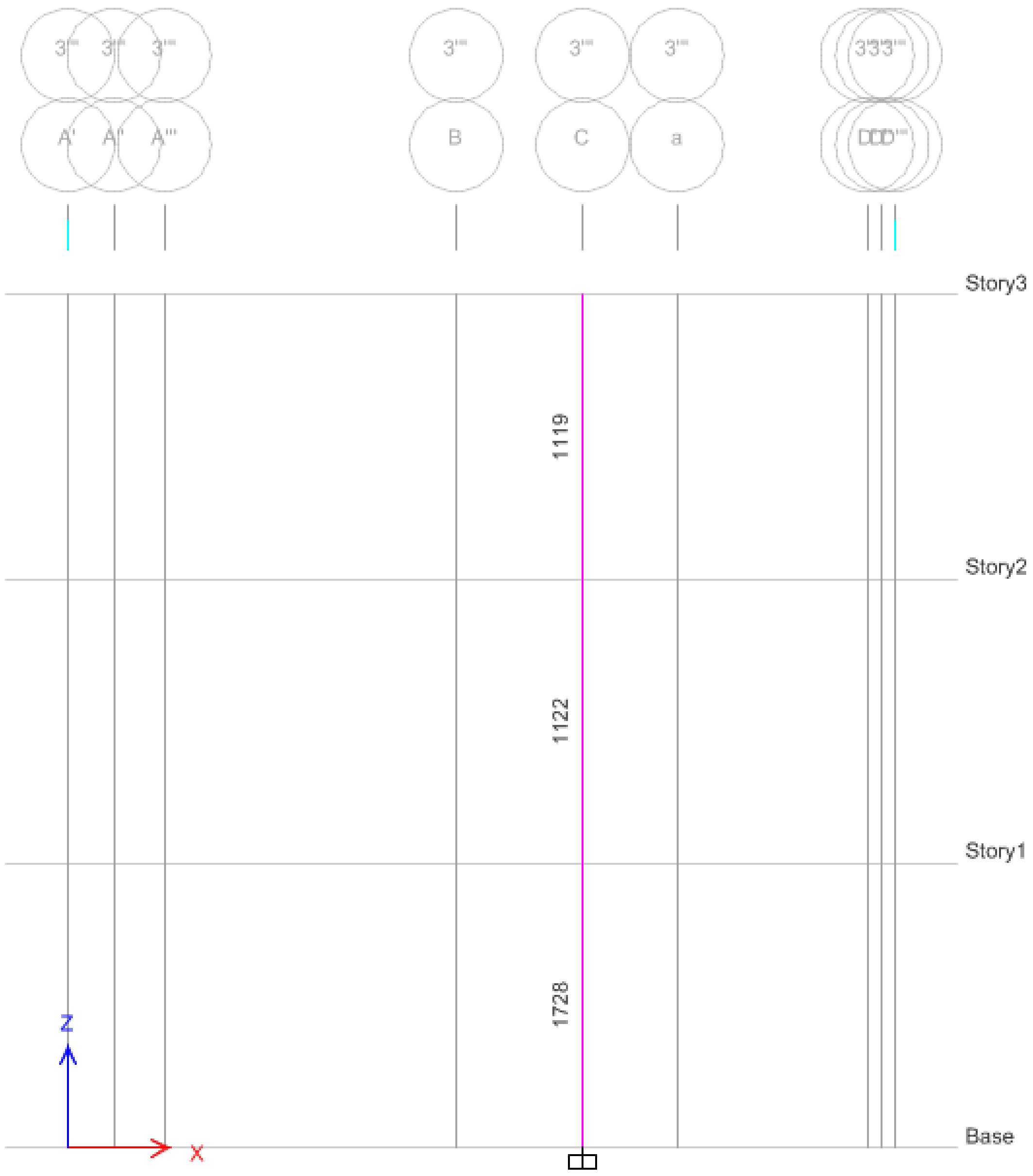
Grid 3-3



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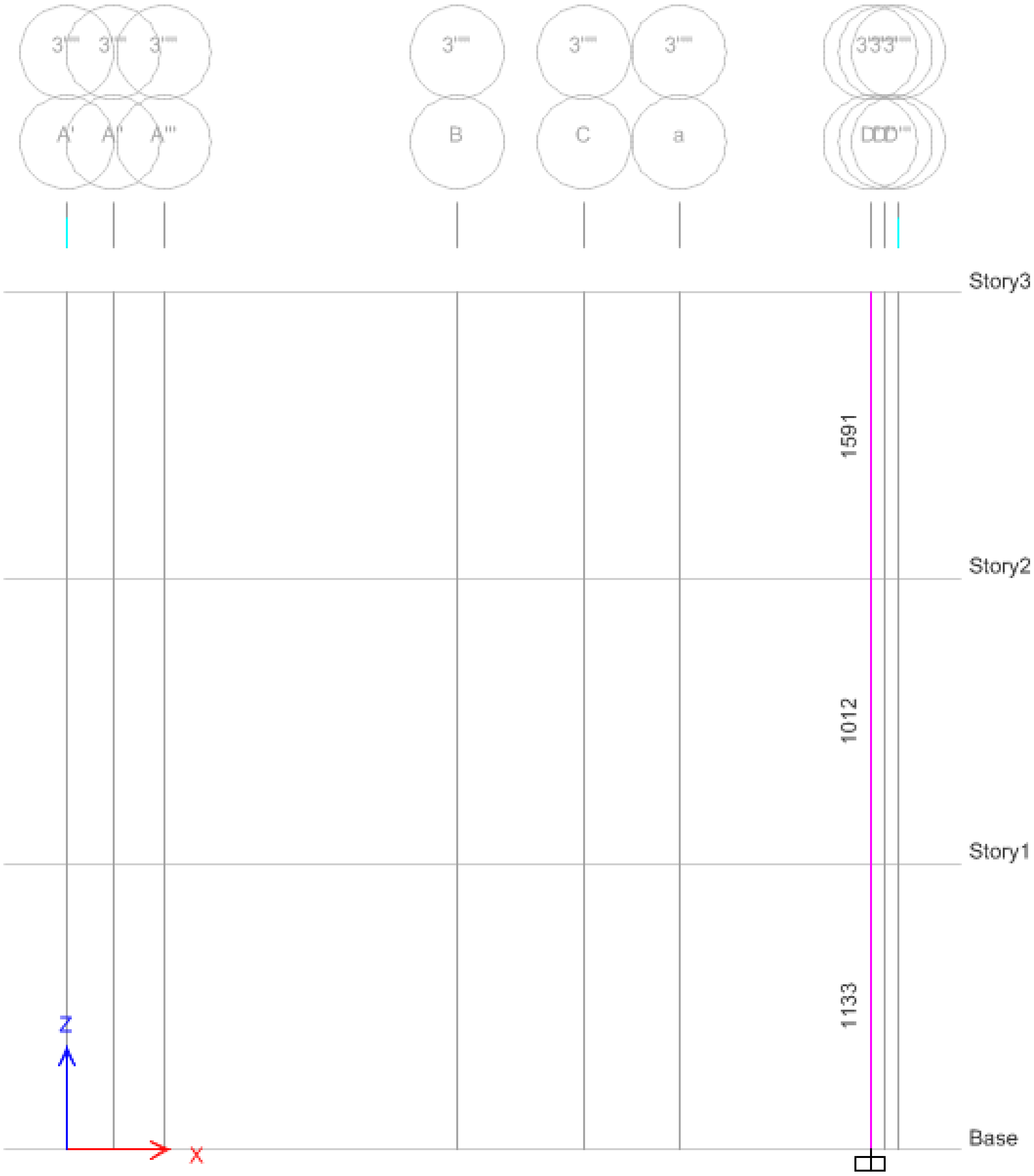
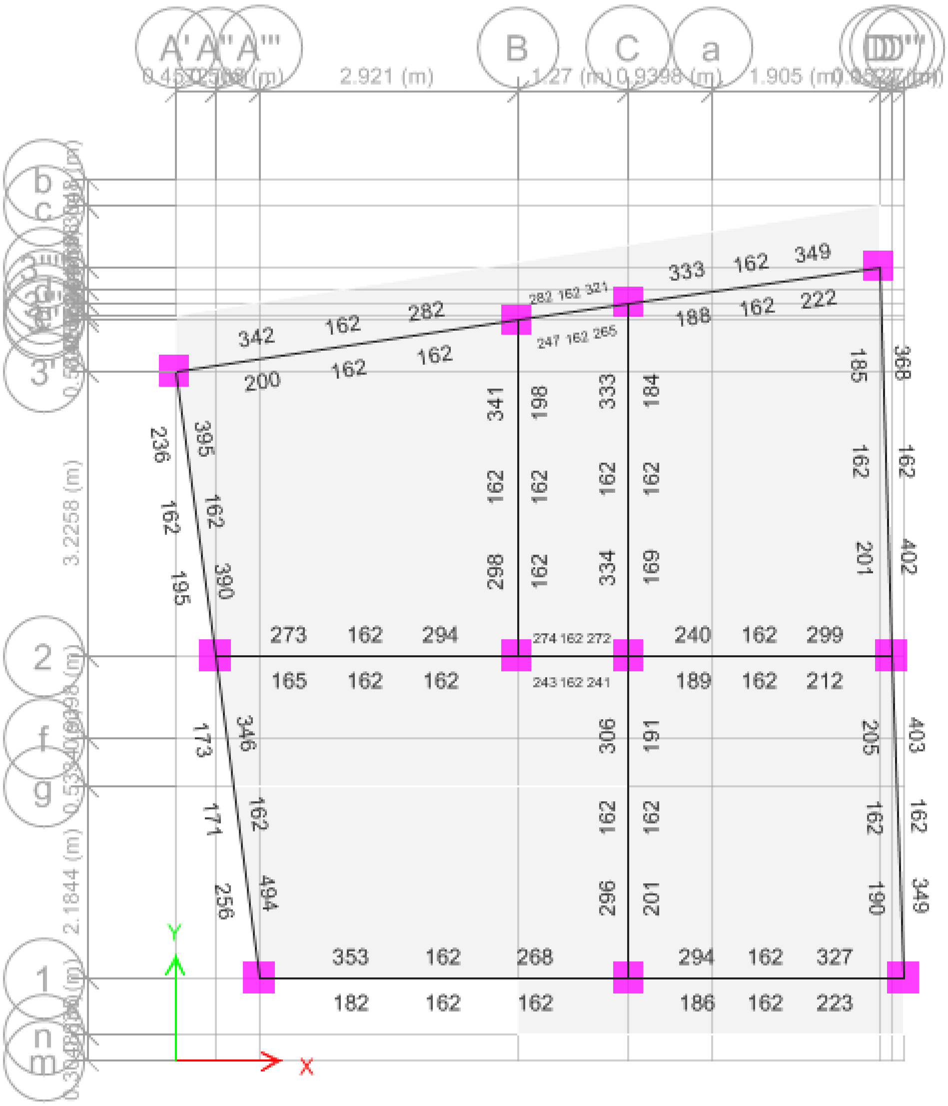


Grid 3-3



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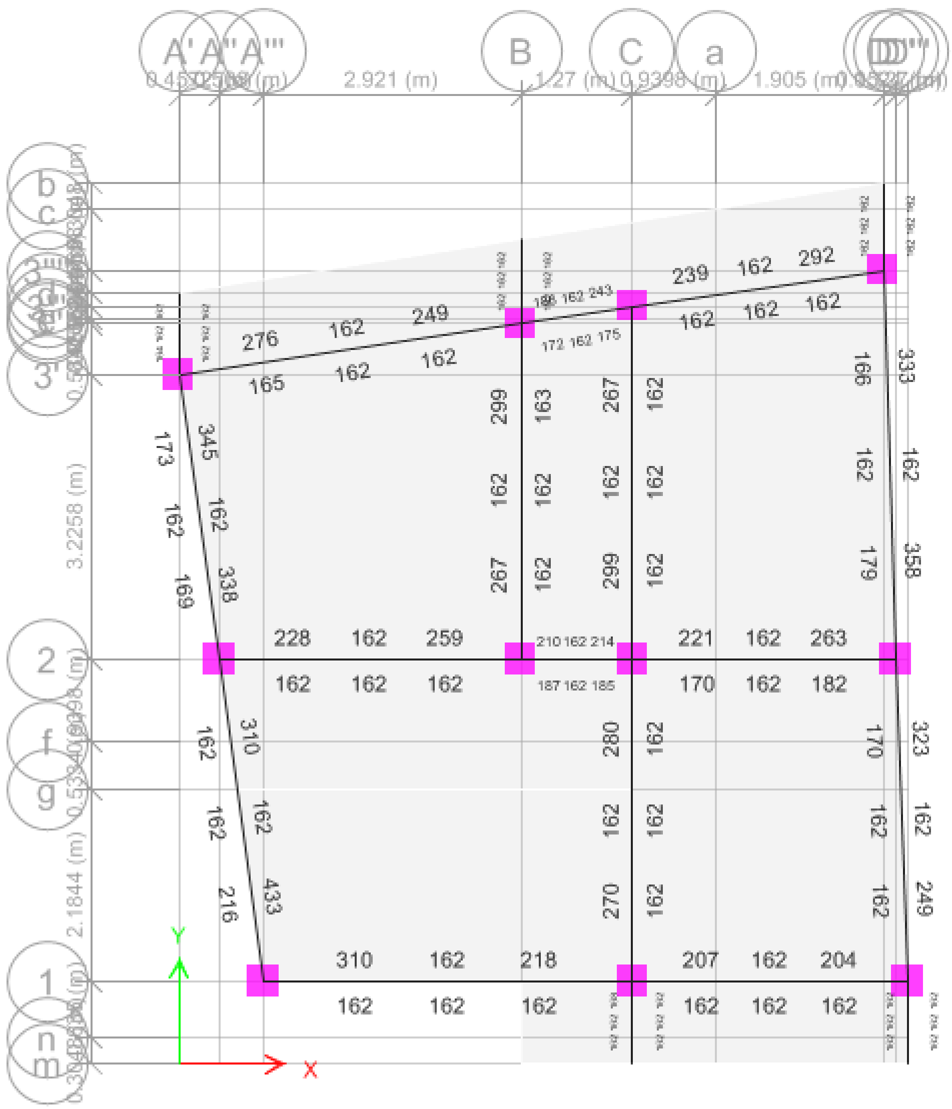
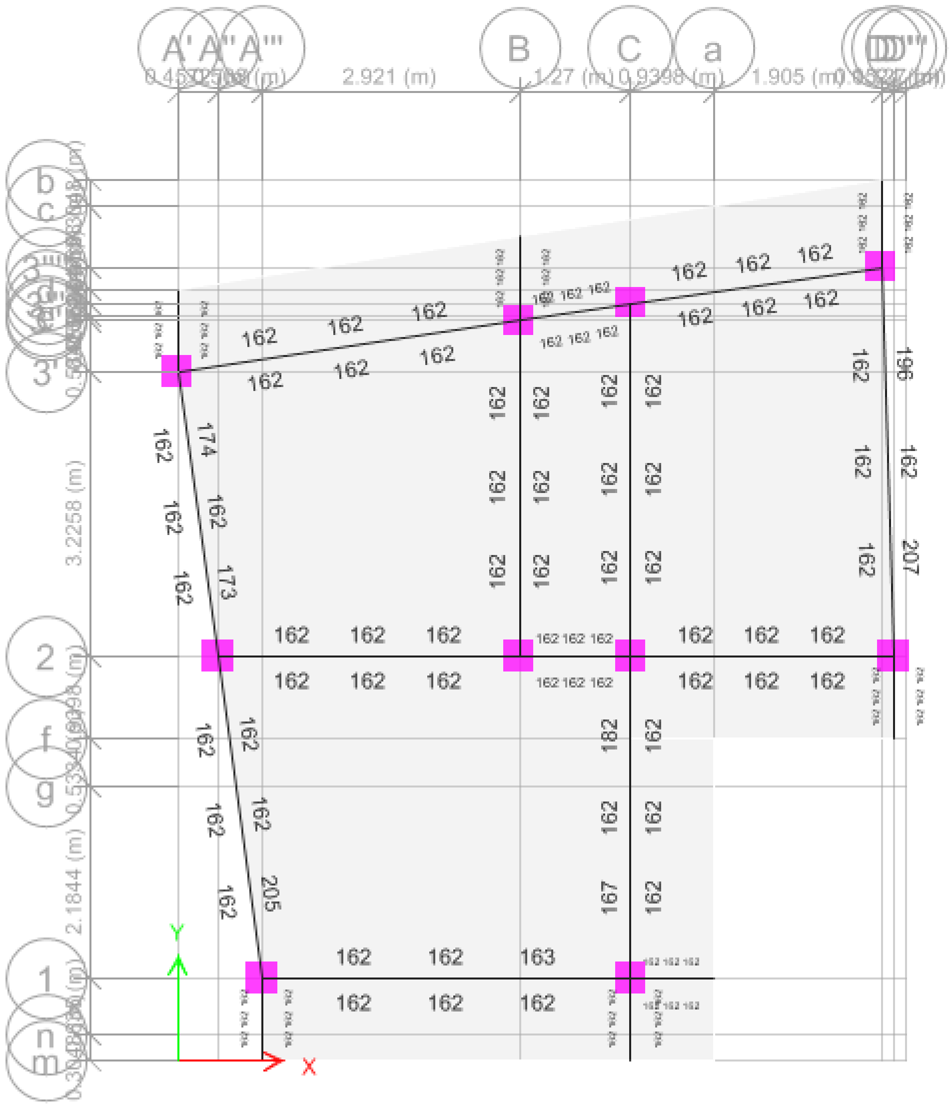
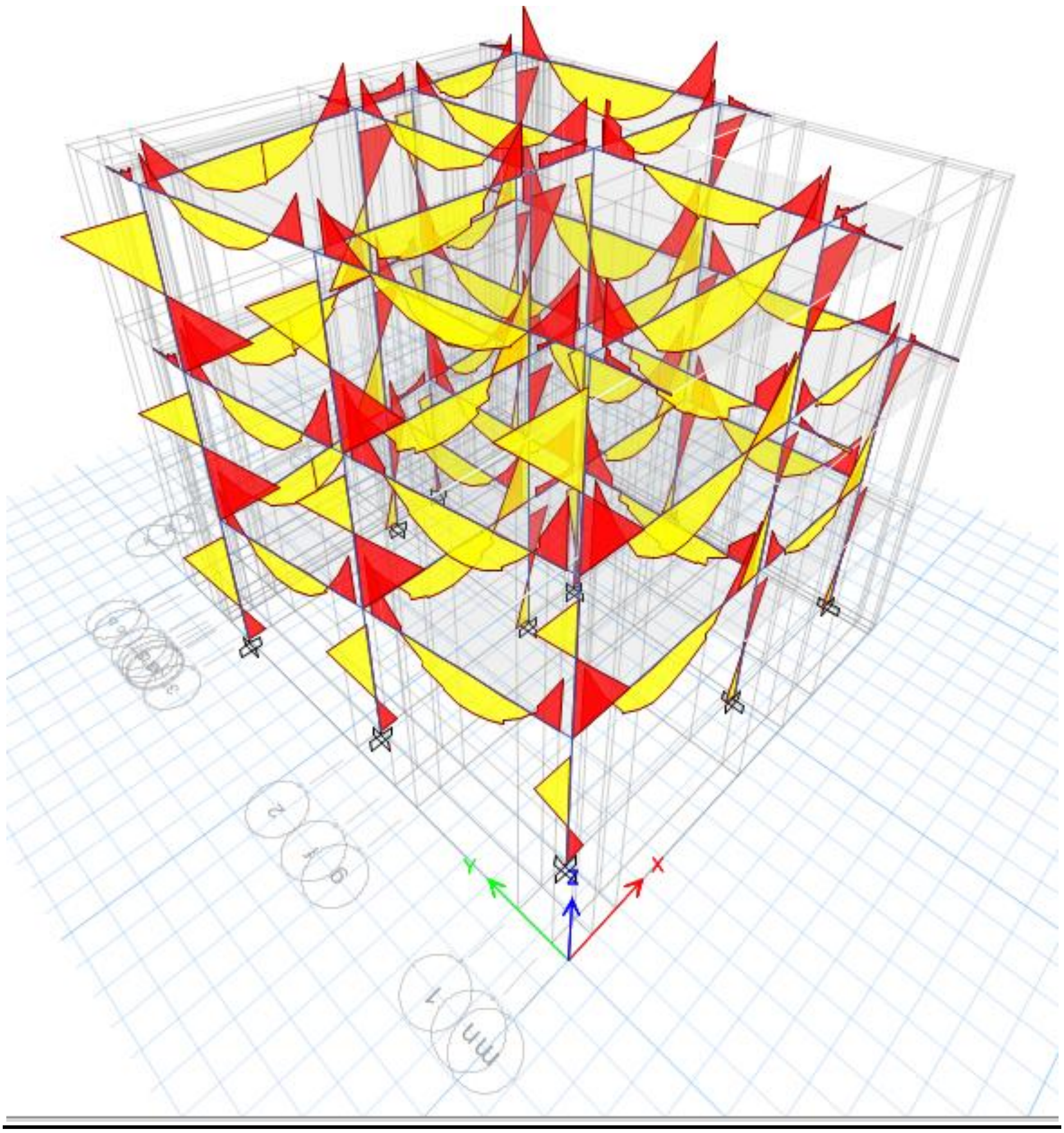
Beam Reinforcements (X -Y Axis) First Floor Beam



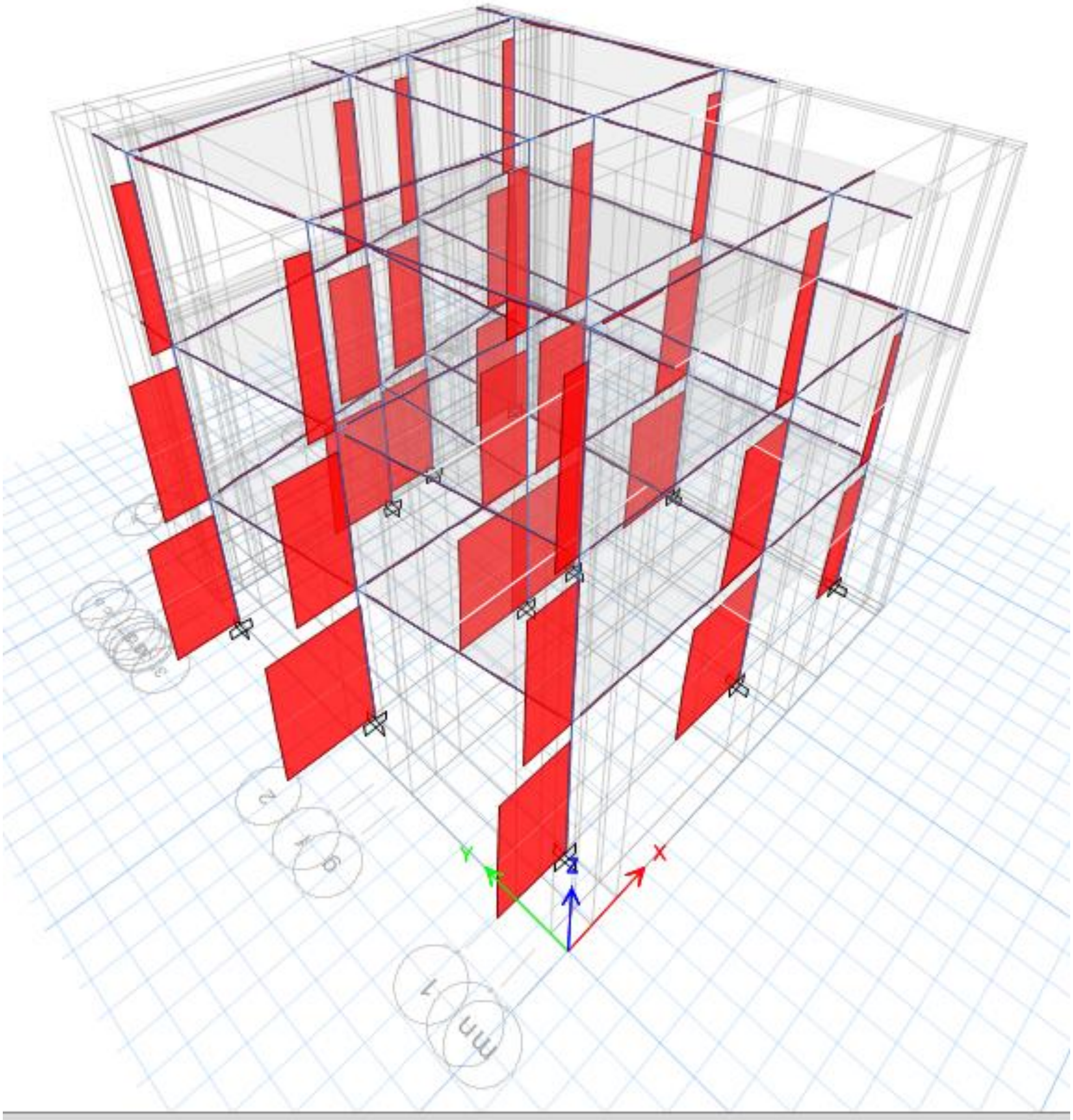
Beam Reinforcements (X -Y Axis) Second Floor Beam



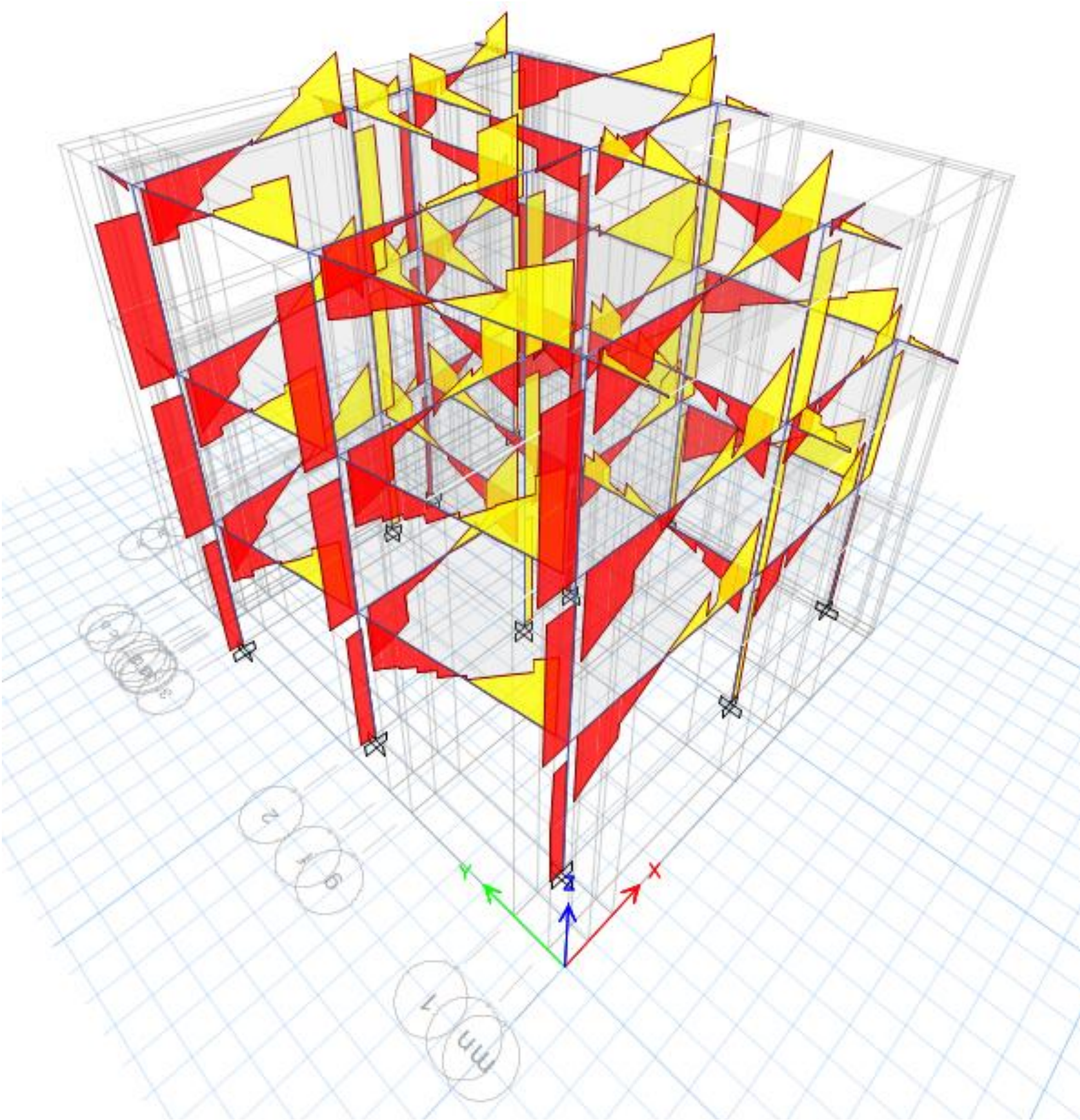
Beam Reinforcements (X -Y Axis) Third Floor Beam



Bending Moment Diagram



Axial Force Diagram



Shear Force Diagram

Seismic Load

Horizontal Base shear coefficient

a) Ultimate Limit State
$C_d(T_1)$ = 0.7875/(4*1.5)
= 0.1313

b) Serviceability Limit State
$C_d(T_1)$= 0.1575/1.25
=0.126

Seismic Weight of the Structure=3871.07KN

Horizontal Seismic Base shear

$V= C_d(T_1)*W$

Where $C_d(T_1)$=Horizontal Base shear coefficient

W=Seismic Weight of the Structure

a) Ultimate Limit State Base Shear

V=0.1313*3871.07

=508.27KN

b) Serviceability Limit State Base Shear

V=0.126*3871.07

=487.75KN

TABLE: Auto Sesmic Load

TABLE: Load Pattern Definitions - Auto Seismic - User Coefficient								
Name	**Is Auto Load**	**Ecc Ratio**	**Top Story**	**Bottom Story**	**C**	**K**	**Weight Used**	**Base Shear**
							kN	kN
EQx SLS	No	0.1	Story3	Base	0.126	1.0655		
EQx SLS(1/6)	Yes	0.1	Story3	Base	0.126	1.0655	2368.6491	298.4498
EQx SLS(2/6)	Yes	0.1	Story3	Base	0.126	1.0655	2368.6491	298.4498
EQx SLS(3/6)	Yes	0.1	Story3	Base	0.126	1.0655	2368.6491	298.4498
EQx SLS(4/6)	Yes	0.1	Story3	Base	0.126	1.0655	2368.6491	298.4498
EQx SLS(5/6)	Yes	0.1	Story3	Base	0.126	1.0655	2368.6491	298.4498
EQx SLS(6/6)	Yes	0.1	Story3	Base	0.126	1.0655	2368.6491	298.4498
EQx ULS	No	0.1	Story3	Base	0.131	1.0655		
EQx ULS(1/3)	Yes	0.1	Story3	Base	0.131	1.0655	2368.6491	310.293
EQx ULS(2/3)	Yes	0.1	Story3	Base	0.131	1.0655	2368.6491	310.293
EQx ULS(3/3)	Yes	0.1	Story3	Base	0.131	1.0655	2368.6491	310.293
EQy SLS	No	0.1	Story3	Base	0.126	1.0655		
EQy SLS(1/3)	Yes	0.1	Story3	Base	0.126	1.0655	2368.6491	298.4498
EQy SLS(2/3)	Yes	0.1	Story3	Base	0.126	1.0655	2368.6491	298.4498
EQy SLS(3/3)	Yes	0.1	Story3	Base	0.126	1.0655	2368.6491	298.4498

TABLE: Modal Participating Mass Ratios								
Case	Mode	Period	UX	UY	UZ	SumUX	SumUY	SumUZ
		sec						
Modal	1	0.605	0.004	0.8389	0	0.004	0.8389	0
Modal	2	0.556	0.8358	0.0046	0	0.8398	0.8436	0
Modal	3	0.518	0.0176	0.0007	0	0.8574	0.8443	0
Modal	4	0.192	0.0001	0.1145	0	0.8575	0.9588	0
Modal	5	0.185	0.0836	0.0013	0	0.941	0.9601	0
Modal	6	0.169	0.0254	0.0017	0	0.9665	0.9618	0
Modal	7	0.112	0.0119	0.0205	0	0.9783	0.9823	0
Modal	8	0.111	0.0168	0.0174	0	0.9951	0.9996	0
Modal	9	0.101	0.0049	0.0004	0	1	1	0
Modal	10	0.018	0	0	0	1	1	0
Modal	11	0.018	0	0	0	1	1	0
Modal	12	0.017	0	0	0	1	1	0

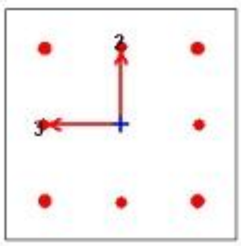
12. DESIGN

12.1 COLUMN DESIGN

Ground Floor= B2

ETABS Concrete Frame Design

IS 456:2000 + IS 13920:2016 Column Section Design (Summary)



Column Element Details

Level	Element	Unique Name	Section ID	Combo ID	Station Loc	Length (mm)	LLRF	Type
Story1	C5	66	Col 14"*14"	DConS19	0	2844.8	1	Ductile Frame

Section Properties

b (mm)	h (mm)	dc (mm)	Cover (Torsion) (mm)
355.6	355.6	60	30

Material Properties

E_c (MPa)	f_{ck} (MPa)	Lt.Wt Factor (Unitless)	f_y (MPa)	f_{ys} (MPa)
22360.68	20	1	500	500

Design Code Parameters

γ_C	γ_S
1.5	1.15

Axial Force and Biaxial Moment Design For P_u , M_{u2} , M_{u3}

Design P_u kN	Design M_{u2} kN-m	Design M_{u3} kN-m	Minimum M_2 kN-m	Minimum M_3 kN-m	Rebar Area mm²	Rebar % %
16.5303	-2.28	88.0464	0.3306	0.3306	1818	1.44

Axial Force and Biaxial Moment Factors

	K Factor Unitless	Length mm	Initial Moment kN-m	Additional Moment kN-m	Minimum Moment kN-m
Major Bend(M3)	0.618627	2489.2	35.7968	0	0.3306
Minor Bend(M2)	0.669084	2489.2	1.0935	0	0.3306

Shear Design for V_{u2} , V_{u3}

	Shear V_u kN	Shear V_c kN	Shear V_s kN	Shear V_p kN	Rebar A_{sv} /s mm²/m
Major, V_{u2}	52.4763	62.971	42.0464	35.7041	394.16
Minor, V_{u3}	19.6676	62.971	42.0464	19.6676	394.16

Joint Shear Check/Design

	Joint Shear Force kN	Shear V_{Top} kN	Shear $V_{u,Tot}$ kN	Shear V_c kN	Joint Area cm²	Shear Ratio Unitless
Major Shear, V_{u2}	N/N	N/N	N/N	N/N	N/N	N/N
Minor Shear, V_{u3}	N/N	N/N	N/N	N/N	N/N	N/N

(1.4) Beam/Column Capacity Ratio

Major Ratio	Minor Ratio
N/N	N/N

Additional Moment Reduction Factor k (IS 39.7.1.1)

A_g cm²	A_{sc} cm²	P_{uz} kN	P_b kN	P_u kN	k Unitless
1264.5	18.2	1819.8075	448.6825	16.5303	1

Additional Moment (IS 39.7.1)

	Consider M_a	Length Factor	Section Depth (mm)	KL/Depth Ratio	KL/Depth Limit	KL/Depth Exceeded	M_a Moment (kN-m)
Major Bending (M_3)	Yes	0.875	355.6	4.33	12	No	0
Minor Bending (M_2)	Yes	0.875	355.6	4.684	12	No	0

Notes:

N/A: Not Applicable

N/C: Not Calculated

N/N: Not Needed

Column Design Summary

SN	Column Name	Column Size		Floor	Reinforcement			Lateral Ties		
		L (in)	B (in)		AR(mm^2)	Bars	AP(mm^2)	Dia./Leg	Side	Mid
1	**A1**	14	14	**GF**	**1403**	8$16	**1608.00**	**8/4L**	**4" c/c**	6" c/c
				1F	**1012**	8$16	**1608.00**	**8/4L**	**4" c/c**	6" c/c
				2F	**1356**	8$16	**1608.00**	**8/4L**	**4" c/c**	6" c/c
				3F						
				4F						
2	**A2**	14	14	**GF**	**1595**	8$16	**1608.00**	**8/4L**	**4" c/c**	6" c/c
				1F	**1081**	8$16	**1608.00**	**8/4L**	**4" c/c**	6" c/c
				2F	**1128**	8$16	**1608.00**	**8/4L**	**4" c/c**	6" c/c
				3F						
				4F						
3	**A3**	14	14	**GF**	**1446**	8$16	**1608.00**	**8/4L**	**4" c/c**	6" c/c
				1F	**1012**	8$16	**1608.00**	**8/4L**	**4" c/c**	6" c/c
				2F	**1315**	8$16	**1608.00**	**8/4L**	**4" c/c**	6" c/c
				3F						
				4F						
4	**B2**	14	14	**GF**	**1818**	4$20+4$16	**2061.00**	**8/4L**	**4" c/c**	6" c/c
				1F	**1212**	8$16	**1608.00**	**8/4L**	**4" c/c**	6" c/c
				2F	**1123**	8$16	**1608.00**	**8/4L**	**4" c/c**	6" c/c
				3F						
				4F						
5	**B3**	14	14	**GF**	**1783**	4$20+4$16	**2061.00**	**8/4L**	**4" c/c**	6" c/c
				1F	**1133**	8$16	**1608.00**	**8/4L**	**4" c/c**	6" c/c
				2F	**1388**	8$16	**1608.00**	**8/4L**	**4" c/c**	6" c/c
				3F						
				4F						
6	**C1**	14	14	**GF**	**1450**	8$16	**1608.00**	**8/4L**	**4" c/c**	6" c/c
				1F	**1012**	8$16	**1608.00**	**8/4L**	**4" c/c**	6" c/c
				2F	**1135**	8$16	**1608.00**	**8/4L**	**4" c/c**	6" c/c
				3F						
				4F						
7	**C2**	14	14	**GF**	**1784**	4$20+4$16	**2061.00**	**8/4L**	**4" c/c**	6" c/c
				1F	**1383**	8$16	**1608.00**	**8/4L**	**4" c/c**	6" c/c
				2F	**1133**	8$16	**1608.00**	**8/4L**	**4" c/c**	6" c/c
				3F						
				4F						
8	**C3**	14	14	**GF**	**1728**	4$20+4$16	**2061.00**	**8/4L**	**4" c/c**	6" c/c
				1F	**1122**	8$16	**1608.00**	**8/4L**	**4" c/c**	6" c/c

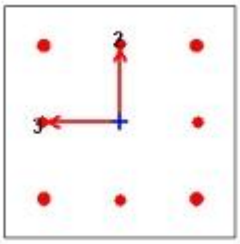
				2F	**1119**	8$16	**1608.00**	**8/4L**	**4" c/c**	**6" c/c**
				3F						
				4F						
9	**D1**	14	14	**GF**	**1396**	8$16	**1608.00**	**8/4L**	**4" c/c**	**6" c/c**
				1F	**1609**	8$16	**1608.00**	**8/4L**	**4" c/c**	**6" c/c**
				2F						
				3F						
				4F						
10	**D2**	14	14	**GF**	**1406**	8$16	**1608.00**	**8/4L**	**4" c/c**	**6" c/c**
				1F	**1020**	8$16	**1608.00**	**8/4L**	**4" c/c**	**6" c/c**
				2F	**1354**	8$16	**1608.00**	**8/4L**	**4" c/c**	**6" c/c**
				3F						
				4F						
11	**D3**	14	14	**GF**	**1133**	8$16	**1608.00**	**8/4L**	**4" c/c**	**6" c/c**
				1F	**1012**	8$16	**1608.00**	**8/4L**	**4" c/c**	**6" c/c**
				2F	**1591**	8$16	**1608.00**	**8/4L**	**4" c/c**	**6" c/c**
				3F						
				4F						

12.2 BEAMDESIGN

Ground Floor = C2-C3

ETABS Concrete Frame Design

IS 456:2000 + IS 13920:2016 Column Section Design (Summary)



Column Element Details

Level	Element	Unique Name	Section ID	Combo ID	Station Loc	Length (mm)	LLRF	Type
Story1	C5	66	Col 14"*14"	DConS19	0	2844.8	1	Ductile Fram

Section Properties

b (mm)	h (mm)	dc (mm)	Cover (Torsion) (mm)
355.6	355.6	60	30

Material Properties

E_c (MPa)	f_{ck} (MPa)	Lt.Wt Factor (Unitless)	f_y (MPa)	f_{ys} (MPa)
22360.68	20	1	500	500

Design Code Parameters

γ_C	γ_S
1.5	1.15

Axial Force and Biaxial Moment Design For P_u , M_{u2} , M_{u3}

Design P_u kN	Design M_{u2} kN-m	Design M_{u3} kN-m	Minimum M_2 kN-m	Minimum M_3 kN-m	Rebar Area mm²	Rebar % %
16.5303	-2.28	88.0464	0.3306	0.3306	1818	1.44

Axial Force and Biaxial Moment Factors

	K Factor Unitless	Length mm	Initial Moment kN-m	Additional Moment kN-m	Minimum Moment kN-m
Major Bend(M3)	0.618627	2489.2	35.7968	0	0.3306
Minor Bend(M2)	0.669084	2489.2	1.0935	0	0.3306

Shear Design for V_{u2} , V_{u3}

	Shear V_u kN	Shear V_c kN	Shear V_s kN	Shear V_p kN	Rebar A_{sv} /s mm²/m
Major, V_{u2}	52.4763	62.971	42.0464	35.7041	394.16
Minor, V_{u3}	19.6676	62.971	42.0464	19.6676	394.16

Joint Shear Check/Design

	Joint Shear Force kN	Shear V_{Top} kN	Shear $V_{u,Tot}$ kN	Shear V_c kN	Joint Area cm²	Shear Ratio Unitless
Major Shear, V_{u2}	N/N	N/N	N/N	N/N	N/N	N/N
Minor Shear, V_{u3}	N/N	N/N	N/N	N/N	N/N	N/N

(1.4) Beam/Column Capacity Ratio

Major Ratio	Minor Ratio
N/N	N/N

Additional Moment Reduction Factor k (IS 39.7.1.1)

A_g cm²	A_{sc} cm²	P_{uz} kN	P_b kN	P_u kN	k Unitless
1264.5	18.2	1819.8075	448.6825	16.5303	1

Additional Moment (IS 39.7.1)

	Consider M_a	Length Factor	Section Depth (mm)	KL/Depth Ratio	KL/Depth Limit	KL/Depth Exceeded	M_a Moment (kN-m
Major Bending (M_3)	Yes	0.875	355.6	4.33	12	No	0
Minor Bending (M_2)	Yes	0.875	355.6	4.684	12	No	0

Notes:
N/A: Not Applicable
N/C: Not Calculated
N/N: Not Needed

Beam Reinforcement Detail

Floor	Size	Main Reinforcement Through		Extra Reinforcement Ends		Stirrup	
		Top	Bottom	Top	Bottom	End	Middle

PlinthTie	12"x12"	2-16φ+1-12 φ	3-16φ			8φ4"c-c	8φ6"c-c
Ground Floor	9"x14"	2-16φ+1-12 φ	3-16φ			8φ4"c-c	8φ6"c-c
1st Floor	9"x14"	2-16φ+1-12 φ	3-16φ			8φ4"c-c	8φ6"c-c
Top Floor	9"x14"	2-16φ+1-12 φ	3-16φ			8φ4"c-c	8φ6"c-c

12.3 SLAB DESIGN

Design of Two way Slab by Limit State Method

(As per IS 456:2000)

1 Data

Longer span of slab, ly		3.531	m
Shorter span of slab, lx		3.506	m
Materials used			
Concrete	M20	20	N/mm2
Steel	Fe500	500	N/mm2

2 Effective span

Longer span of slab, ly	3.354	m
Shorter span of slab, lx	3.329	m
ly/lx	1.008	
type of slab	<2 two way slab	

Support Condition :

Two Adjacent Edge Discontinuous

3 Calculation of loads

i) Dead load

i	Slab thickness	125	mm
	Self weight of concrete	25	KN/m3
	Dead load of slab = 25 * 0.125	3.125	KN/m2
ii	Screeding thickness	30	mm
	Self weight of screeding	20.4	KN/m3
	Dead load of screeding = 20.4 * 0.03	0.612	KN/m2
iii	marble flooring thickness	19	mm
	Self weight of marble	27	KN/m3
	Dead load of marble = 27 * 0.019	0.513	KN/m2
iv	Cement plaster thickness	12.5	mm
	Self weight of plaster	20.4	KN/m3
	Dead load of plaster = 20.4 * 0.0125	0.255	KN/m2

Total dead load on slab	**4.505**	**KN/m2**

ii) Live load

For Residential Building	2	KN/m2
Total design load (DL+LL)	**6.505**	**KN/m2**
Factored load = 1.5 * design load	9.7575	KN/m2

4 Calculation of moments

As per code,

-ve moment at support

+ve moment at mid

Shorter Direction		Longer Direction	
Support, αx1	0.047	Support, αy1	0.047
mid, αx2	0.035	mid, αy2	0.035

Along shorter span,		
Mx = αx1*wu*lx	6.407	KNm
My = αy1*wu*lx	6.346	KNm
Along longer span,		
Mx = αx2*wu*lx	4.777	KNm
My = αy2*wu*lx	4.726	KNm

5 Depth Calculation

Minimum depth of slab is calculated considering maximum bending moment

Maxm bending moment 6.407

Depth required,

d=sqrt(Mu/(0.36*xumax/d)*(1-0.42*xumax/d)*b*fck)

overall depth, D=d+φ/2+clear cover)		mm
For safe design, adopting depth of slab	125	mm
Effective depth	96	mm

6 Reinforcement Calculation

	Main Bar	Dist. Bar	
provide the reinforcement of dia	10	8	mm

S. No.	Description	Shorter Direction		Longer Direction	
		Support	mid	Support	mid
1	αx & αy	0.047	0.035	0.047	0.035
2	BM	6.41	4.78	6.35	4.73
3	Ast required	145.30	107.32	143.87	106.15
4	Ast min	150.00	150.00	150.00	150.00
5	Spacing reqd	523.81	523.81	335.24	335.24
6	Spacing prov.	150.00	150.00	150.00	150.00

7	Ast provided	523.81	523.81	335.24	335.24
8	% of steel provided	0.42	0.42	0.27	0.27

7 Check

7.1 Minimum reinforcement

Min Ast = 0.12% bD 150 mm2 < Ast provided in all sections

7.2 Check for deflection

From deflection control criteria

lx/d < (l/d)basic*αβγ where

lx/d 31.7

α = modification factor for tension reinforcement

β = modification factor for compression reinf. 1

γ = reduction factor for ratios of span to eff. depth in flanged beam 1

(l/d)basic = 23 of IS 456

For α,

fs = 0.58*fy*(Ast required/Ast provided) 83.045 N/mm2

% of tensional reinforcement 0.419

α 2

now, (l/d)basic*αβγ 46 OK Safe

Since, lx/d < (l/d)basic*αβγ SAFE

7.3 Check for Shear

Shear is checked along short span.

Maximum shear force, Vumax = w*lx/2 20.283 KN

Normal shear stress, τv = V/bd 0.193 N/mm2

% of steel provided = 100*Ast provided/bD 0.249

Shear capacity of section without reinforcement

From table 19,

τc 0.360 N/mm2

Shear strength, τ'c = k*τc

For slab thickness 125mm, k 1.3

Shear strength, τ'c 0.467 N/mm2

Since, τ'c > τv OK Safe

8 Results

Overall depth of slab =	125	mm
Main bar along short span at support	10mm ø @ 150mm c/c	
Main bar along short span at mid	10mm ø @ 150mm c/c	
Distribution Bar along long span at support	8mm ø @ 150mm c/c	
Distribution bar along long span at mid	8mm ø @ 150mm c/c	

12.4. Design of Staircase

Y =		25	N/m3
Effective Span		3.3	m
Riser (R)		152	mm
Tread (T)		228	mm
$\sqrt{(R^2+T^2)}$		310.16	mm
Assume waist slab thickness (d)		100	mm
	(D)	125	mm
Assumed grade of concrete (fck)		M20	
Assumed steel (fy)		Fe 500	

Loads on stairs

Self weight of waist slab	3 82	KN/m2
Self weight o steps	2.225	KN/ m^2
Finish	1	KN/m^2
Live Load	3	KN/m^2
Total Load	10.04	KN/m^2

Loads on landing

Self weight of slab	3.125	KN/m^2
Finish	1	KN/m^2
Live Load	3	KN/m^2
Total Load	7.125	KN/m^2

Design Moment considering 1 m wide strip fo waist slab.

Reactions

R_A =	18.09	KN
R_B =	12.92	KN

Max moment at x=	1.580	m from A support
BMmax	11.70	KNm/m

Depth required

M=2.66bd^2 (for M20 and Fe500)

d	66.32	mm
d (provided)	120	mm (OK)

Ast required

BM= 0.87fyAst (d-(Ast*fy)/(fck*b))

Ast	235.71	mm2
φ =	12	mm
spacing	480.00	mm

<u>Distributors</u>

= 0.12% of b D	150	mm2
φ =	8	mm
spacing	335.24	mm

12.4
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<u>Shear check</u>

Max shear at distance d from the support

V_u =	17.87	KN
$T_v = V_u/b\ d$	0.179	N/mm

T_c

% = 100 * (Ast/b d)	0.235713066	%
T_c =	0.44	for M 20
k =	1.3	
k T_c =	0.57	N/mm2>Tv (OK)

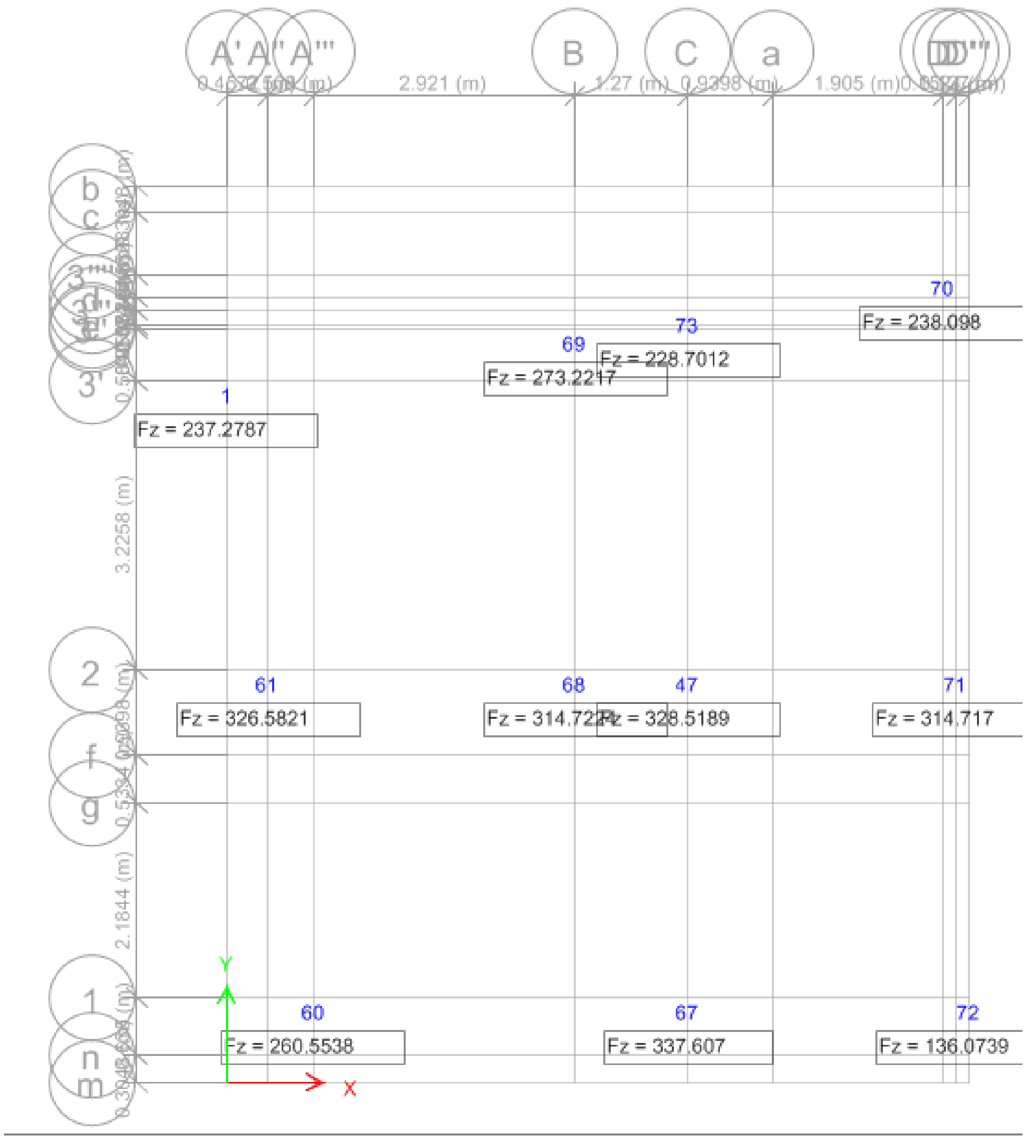
Provide 12 mm φ @ 150 mm spacing c/c **Main bars**
Provide 10 mm φ @ 150 mm spacing c/c **Distribution bars**

12.5. Design of foundation

The purpose of the foundation is to effectively support the superstructure by transmitting the applied load effects to the soil below, without exceeding the safe bearing capacity of the superstructure by ensuring settlement of the structure is within tolerable limits, and as nearly uniform as possible.

The choice of the type of the foundation depends not only on the type of the superstructure and the magnitude and types of reactions induced at the base of the superstructure, but also on the nature of the soil strata on top of which the substructure is to be founded.

The foundation used for this building is isolated foundation. Bearing Capacity of Soil is 150KN/m2.



Showing Joint Reaction along respective joint:

Isolated Footing Design

Foundation Type :	**F2**			
Required Data				
size of colunm=				
	l =	0.35	m	
	b =	0.35	m	
Bearing Capacity of Soil=		120	KN/m²	
Strength of Steel (fy)=		500	N/mm²	
Strength of Concrete (fck)=		20	N/mm²	(M 20)
Factor Load=		337.608		
Axial Load =		225.07	KN	
Approximate area of		1.88	m²	

footing=						
Weight of footing including earth=			56.27			
Axial Load =			225.07	KN		
Total weight on soil=			281.34	KN		
Actual Area=			2.34	m²		
Size of Square Footing =			1.531			5.02225734
Adopted size=						
	L =		1.83	m		6.00
	B =		1.83	m		
Actual area of footing =			3.3489	m²		
Net Pressure acting upwads =			100.81	KN/m²		
B. M. at the face of column about an axis =				50.51		KN-m
The effective depth required is given by						
			BM= 0.138 fck*bd^2			
	d=		100.00	mm		
Adopt d = 2 to 3 Times of calculated value of d for Shear considerations.						
	Adop. d=		300	mm		
	D=		350	mm		
<u>**Check for one-way Shear action**</u>						
The critical section is taken at distance d away from the face of column						
Shear force Vu=		81.17	KN			
Nominal shear stress (Tv) =			Vu/b*d =	0.15	N/mm²	
Allowable Shear Stress of Concrete for pt% of Steel < = 0.15 &					(M 20)	0.28
For 0.165%	Tc =	0.292	N/sq.mm	(From		

				IS 456: 2000, table 19)
		Hence,	OK	
	Tc should be equal to or greater than Tv			

Check for two-way Shear action

The critical section is taken at a distance of 0.5d away from the face of column ,

	Shear force Vu=			295.02	KN
Nominal Shear Stress(Tv) = Vu/b'd (Where b' is the periphery of critical section)					
	and		b' =	2600	mm
So, Nominal Shear Stress(Tv) =				0.38	N/mm²
Shear Strength of concrete is					
Tc =	1.12	N/sq.mm			
T'c =	Ks*Tc	Where Ks=(0.5+Bc) Bc=length of shorter side of column/length of longer side of column			
		Bc=		1	
		Ks=		1.5	should not be greater than 1
		Take(Ks)=	1		
		T'c=		1.12	N/mm²
		Hence,	OK		T'c should be greater than Tv

Area of Steel Calculation,

Area of Steel is given by formula, BM = 0.87* fy* At*{d- fy*At/(fck*b)}

Then,

Ast =	394.14	sq.mm	> Ast min	= 0.12% of b * d	
Ast per m =	215.38			360	mm^2
Req. Area of steel =		360.00	mm^2		
% of steel=		0.072			

Steel requied as	12	mm bar @	314.16	mm c/c distance.
Provide Steel as =	12	mm bar @	150	mm c/c distance.

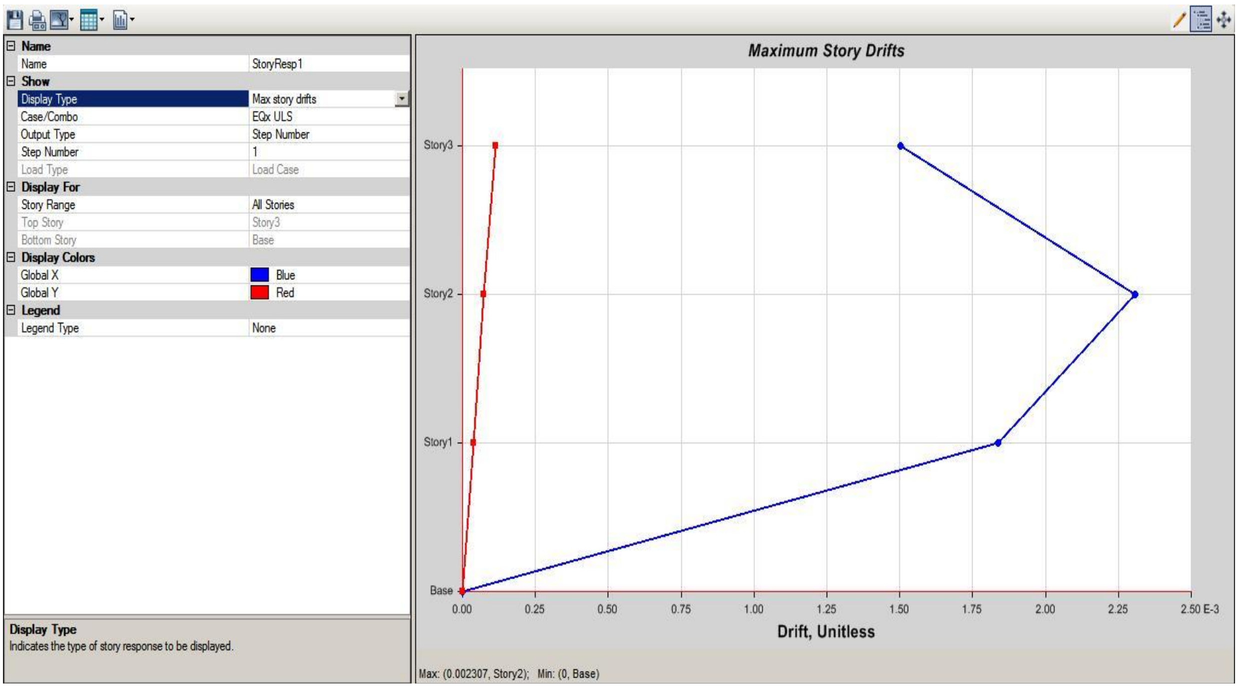
Provided steel = 753.98 mm^2 Hence Ok.

The output of the design of footing is presented below. The construction shall follow the detail drawing of footing.

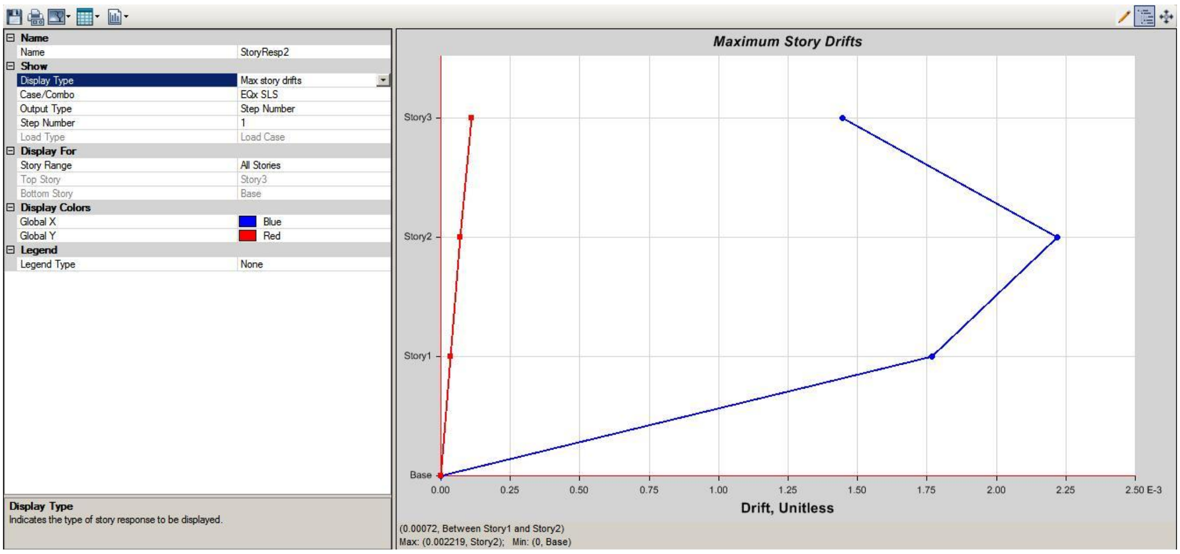
Node No.	Footing	Footing Size	Rebar	Thickness
A1, D1, A3, D3	F1	5'0"X5'0"	Φ12@ 150mm c/c both direction	1'-10"
C1, A2, D2	F2	6'0"X6'0"	Φ12@ 150mm c/c both direction	1'-10"
C2 & C3, B2 & C2	CF	11'-2"X7'0"	Φ12@ 150mm c/c both direction	1'-10"

13. ACCEPTABILITY CRITERIA

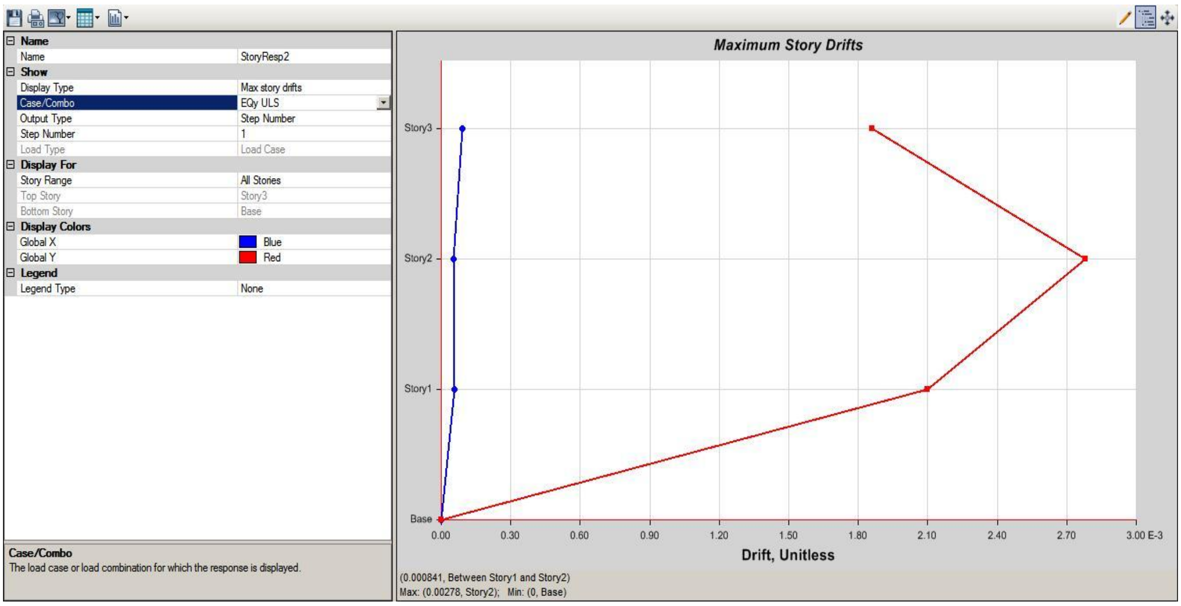
13.1 STORY DRIFT



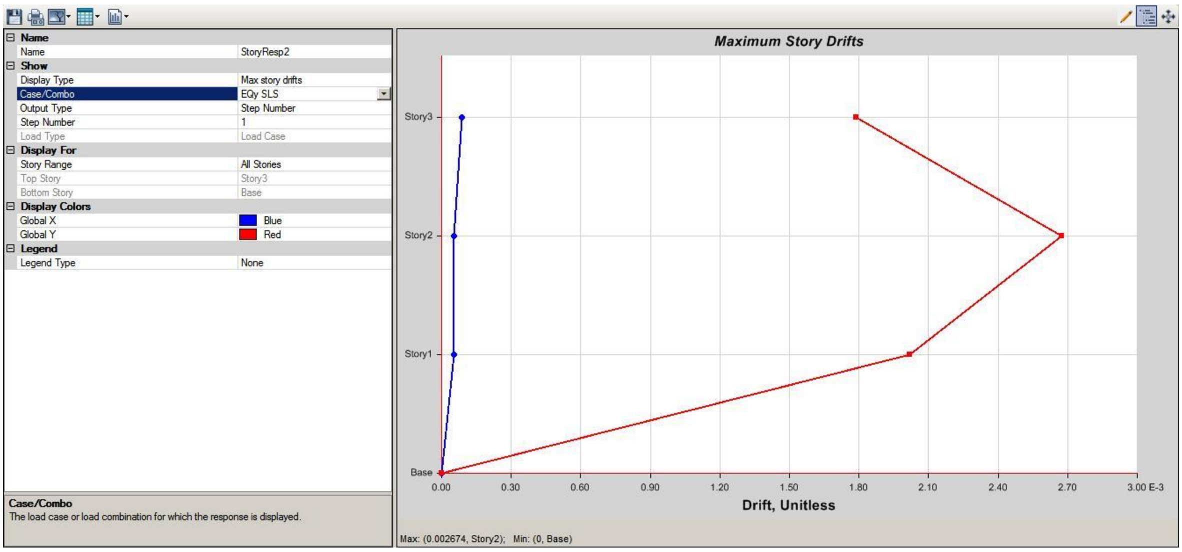
Inter-storeyDrift(Load case:ULSEQx)



Inter-storeyDrift(Load case:ULSEQy)



Inter-storeyDrift(Load case:SLSEQx)



Inter-storeyDrift(Load case:SLSEQy)

To control overall deflection, the criteria given in clause 5.6.1 of NBC 105:2020 is applied. The maximum design horizontal deflection in any story due to the design lateral force shall be determined by multiplying the horizontal deflection found from

Equivalent Static Method or Modal Response Spectrum Method by the Ductility factor (Rμ). For ultimate limit state and for serviceability limit state it shall be taken as equal to the horizontal deflections calculated either by Equivalent Static Method or Modal Response Spectrum Methods.

Furthermore, the drift shall not exceed maximum drift in the model. The maximum drift for ultimate limit state is 0.005203*4=0.0208 which is less than 0.025. The maximum drift for serviceability limit state is 0.004993 which is less than 0.006.

14. DESIGN INPUT IN MUNICIPALITY FORMAT

A. Geometrical Configuration ofStructure:
- (i) No. of Blocks:-1
- (ii) No. of Storey:-2.5
- (iii) Total Height of the Structure:-9.433m

B. Structural Analysis andDesign:
- (i) Structural Analysis Software:- ETABS2020
- (ii) Structural System: - RC buildings with Special Moment Resisting Frame SMRF.
- (iii) Foundation System:-Isolated, Combined and Strap Footing
- (iv) Loading Parameters:
  - a) Dead Load:- As per IS875:1987 PartI
  - b) Live Load:- As per IS875:1987 PartII
  - c) Seismic Load (As perNBC 105:2020):-
- (v) Maximum Deflection of theBuilding:-
- (vi) Maximum Drift in theBuilding:-
- (vii) Load Combination Considered:-NBC 105:2020
- (viii) Concrete Design code Referred:- IS456:2000
- (ix) Ductile Detailing codeReferred:- IS13920:2016
- (x) Concrete Grade Used:
  Beam –M20
  Slab – M20
  Column – M25
  Foundation – M20
- (xi) Reinforcement Grade Used: -Fe500